

Physical Chemistry (I)

Lecture 20

Special Lecture:

Statistical Mechanics of Proteins



Phase Transitions

Ordered phase

energetically
favored

Interactions



Disordered phase

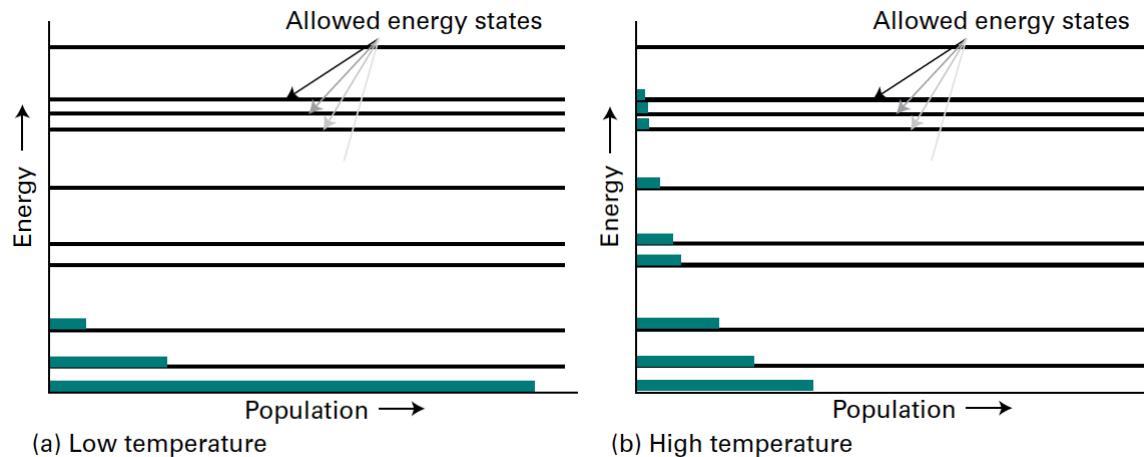
entropically
favored

$$S = k_B \ln W$$

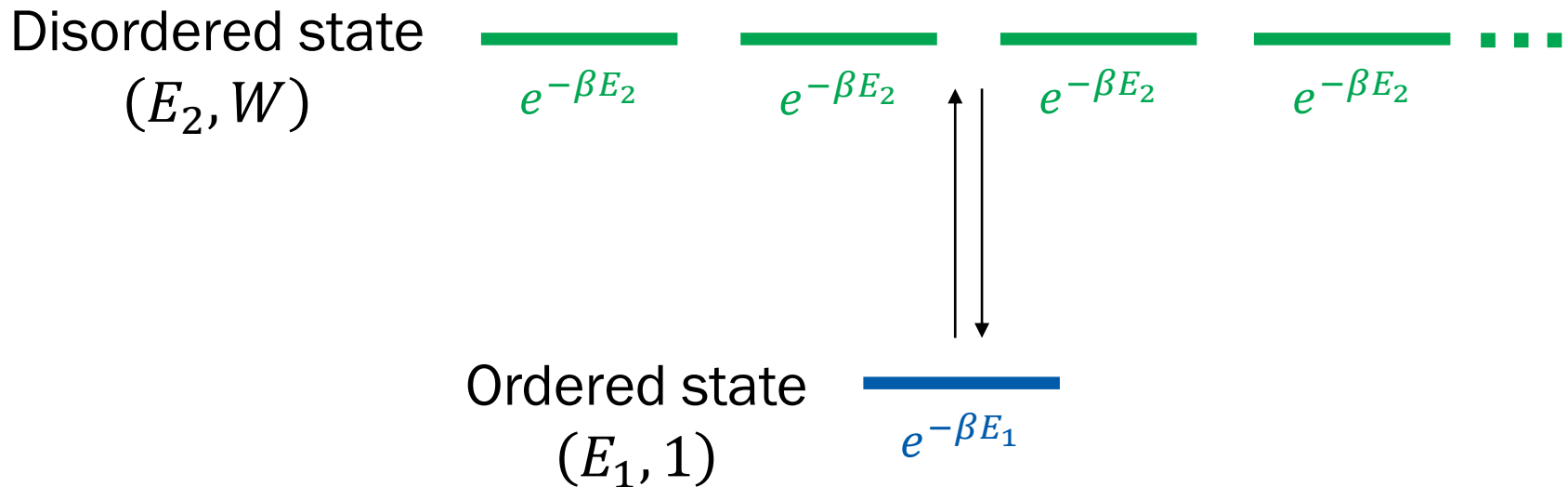
Boltzmann Distribution

$$\text{Probability} \propto e^{-E/k_{\text{B}}T} = e^{-\beta E}$$

(at thermal equilibrium)



Two-State Model

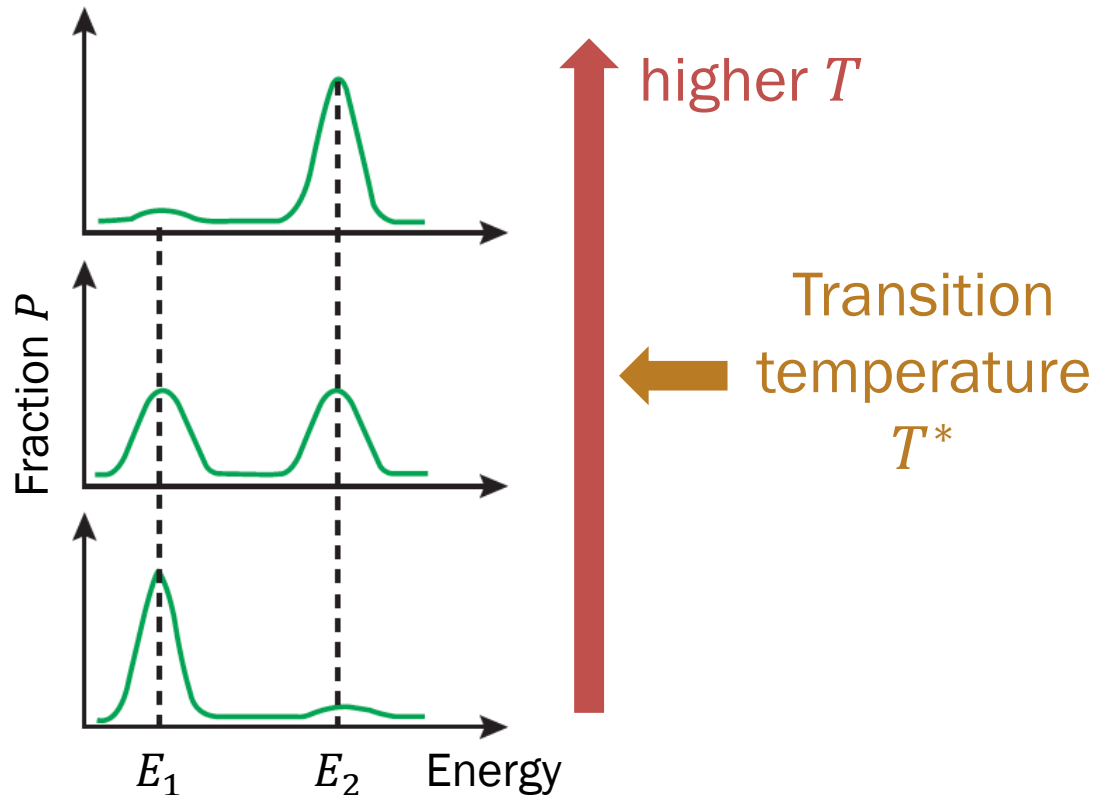


$$P_1 = \frac{e^{-\beta E_1}}{e^{-\beta E_1} + W e^{-\beta E_2}}$$

Phase Transition

$$P_1 = \frac{e^{-\beta E_1}}{e^{-\beta E_1} + W e^{-\beta E_2}}$$

$$P_2 = \frac{W e^{-\beta E_2}}{e^{-\beta E_1} + W e^{-\beta E_2}}$$



Transition Temperature

$$P_1 = \frac{e^{-\beta E_1}}{e^{-\beta E_1} + W e^{-\beta E_2}}, \quad P_2 = \frac{W e^{-\beta E_2}}{e^{-\beta E_1} + W e^{-\beta E_2}}$$

At the transition temperature, $P_1 = P_2$

$$e^{-\beta E_1} = W e^{-\beta E_2} = e^{-\beta(E_2 - T^* S_2)}$$

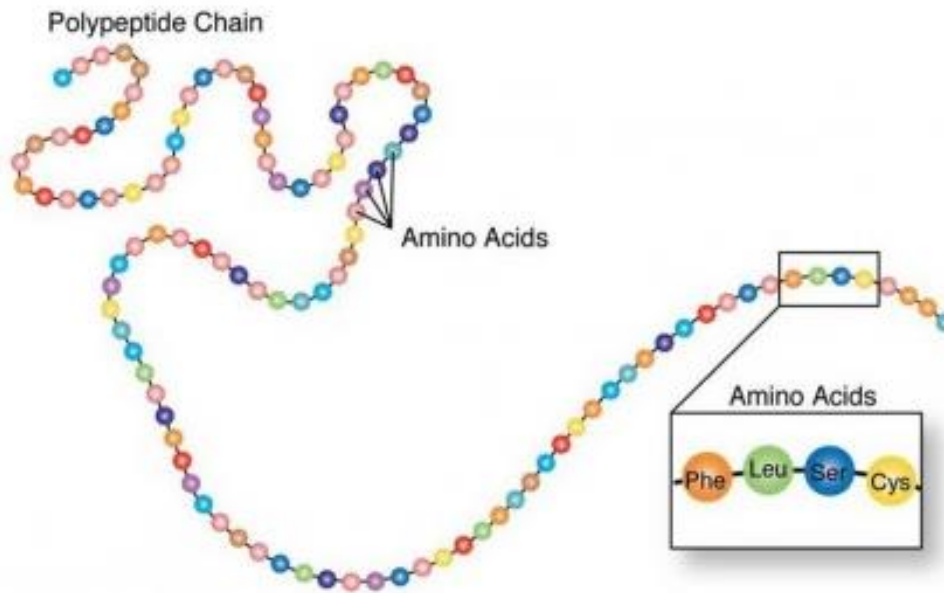
Hence,

$$E_1 = E_2 - T^* S_2$$
$$T^* = \frac{E_2 - E_1}{S_2} = \frac{E_2 - E_1}{k_B \ln W}$$

Now Let's Talk About **Proteins** ...



Proteins are Polymer Chains



Proteins are Polymer Chains

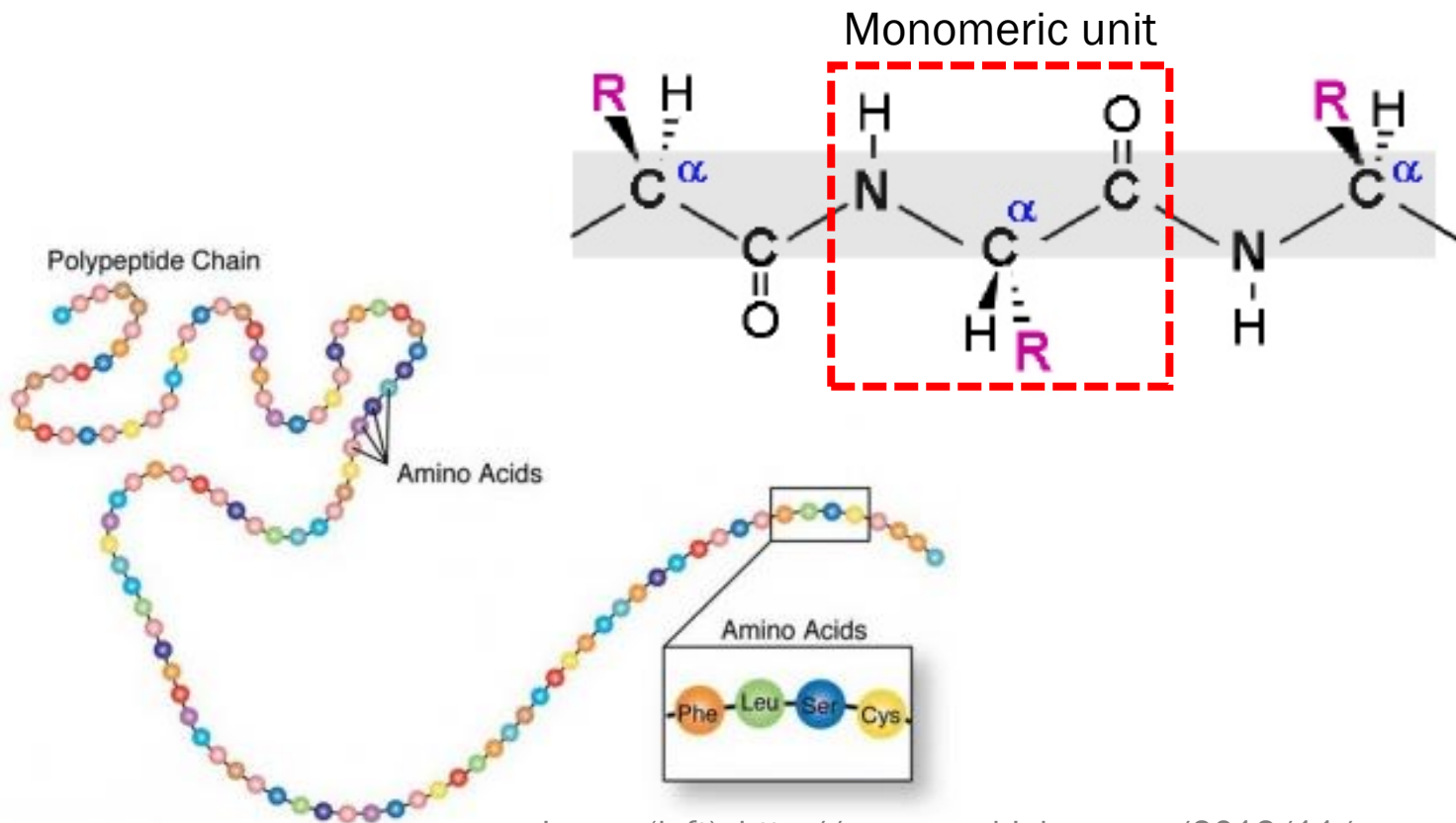


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

Proteins are Polymer Chains

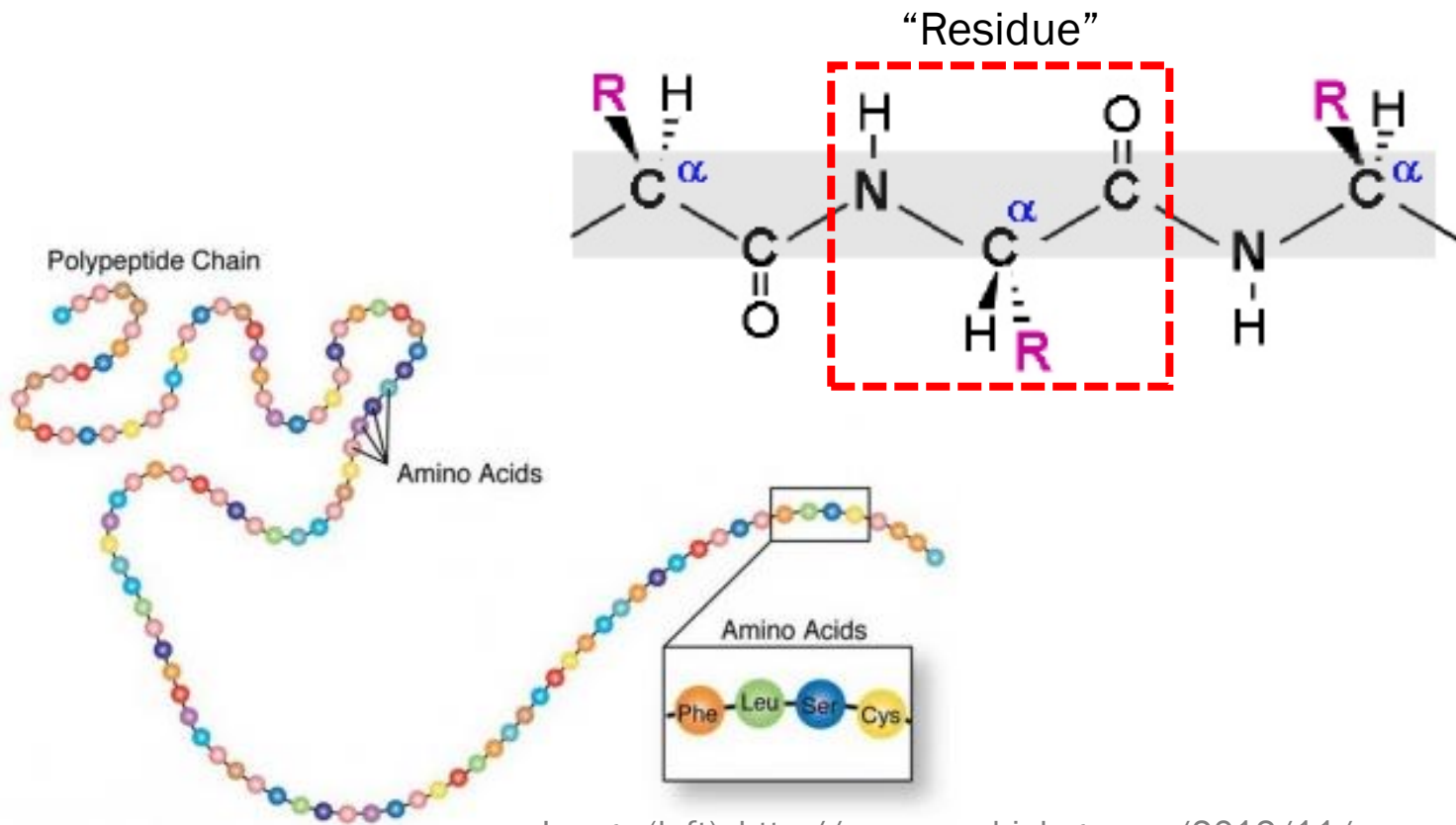


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

Proteins are Polymer Chains

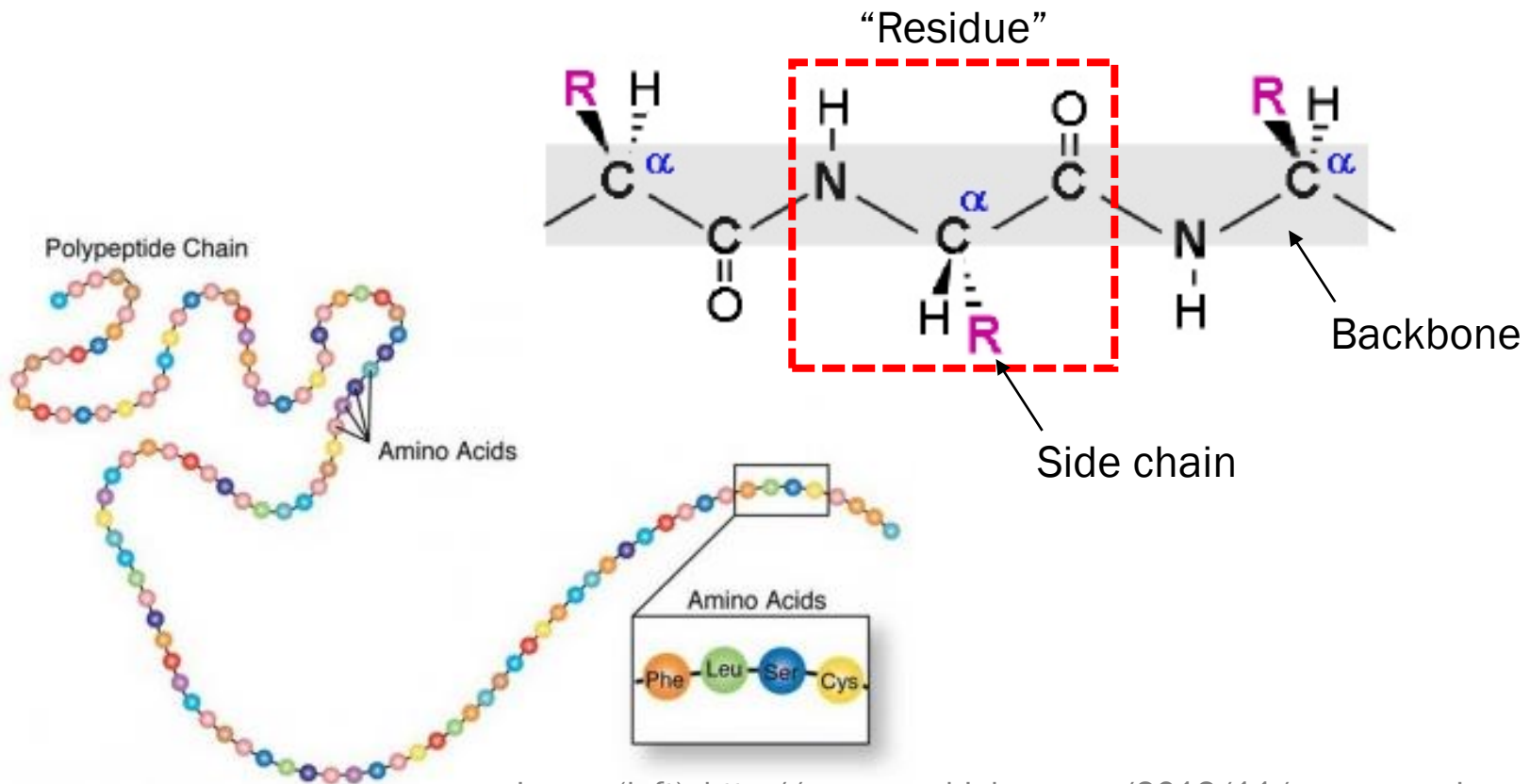


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

20 Typical Monomeric Units

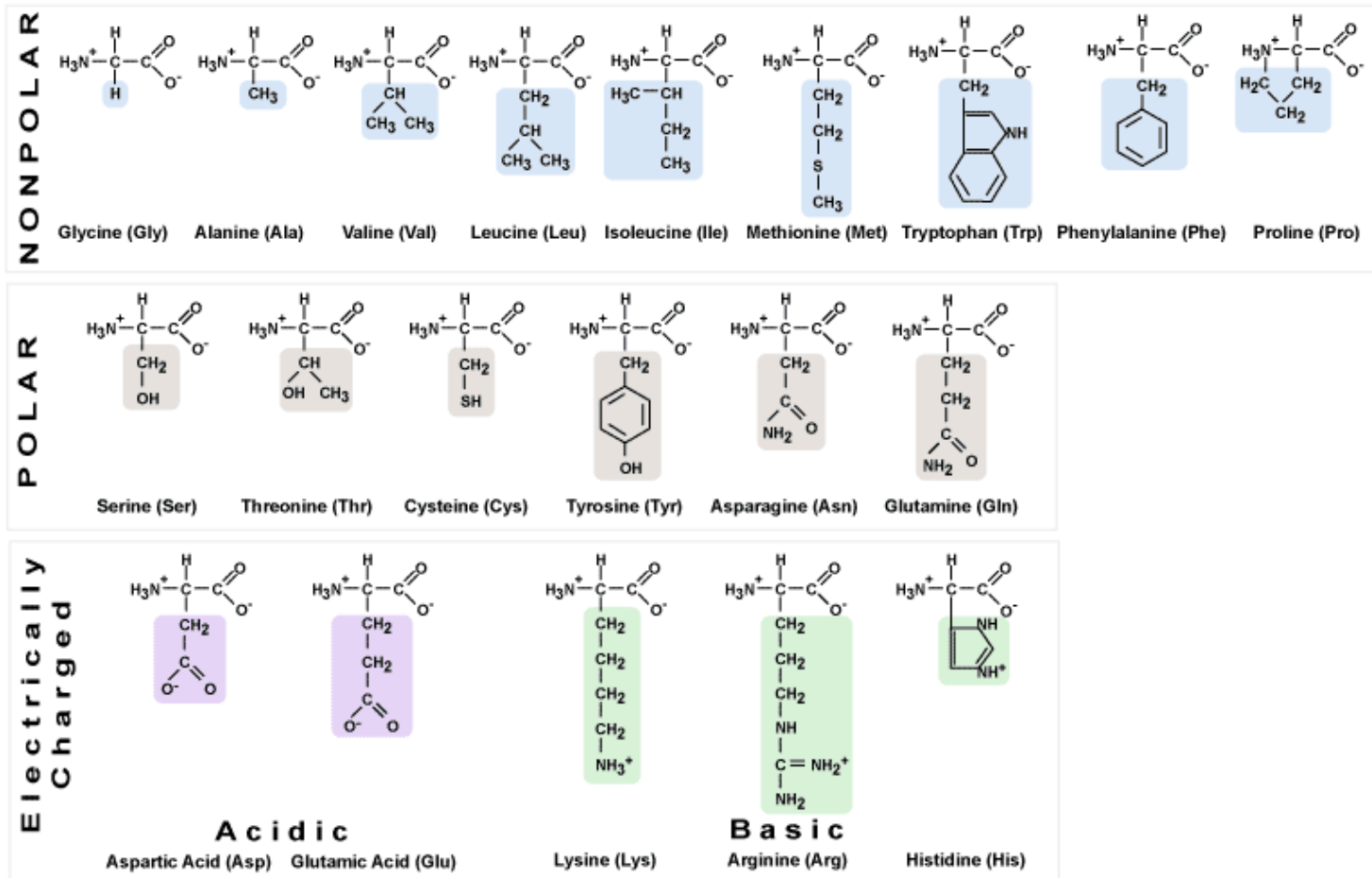


Image: <https://nuraisyahsukor.weebly.com/class-of-amino-acid-nitrogen-cycle-urea-cycle.html>

Proteins are Polymer Chains

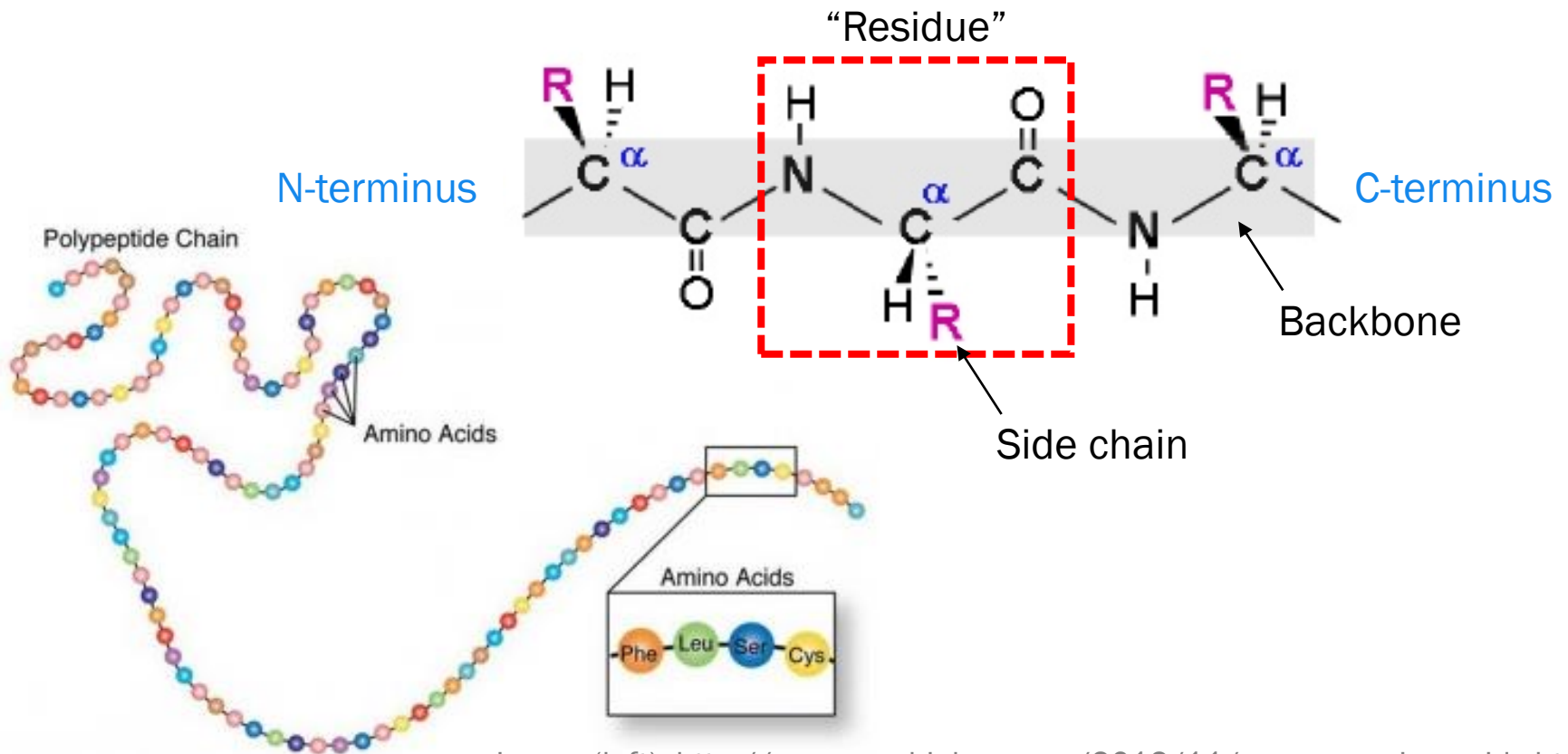
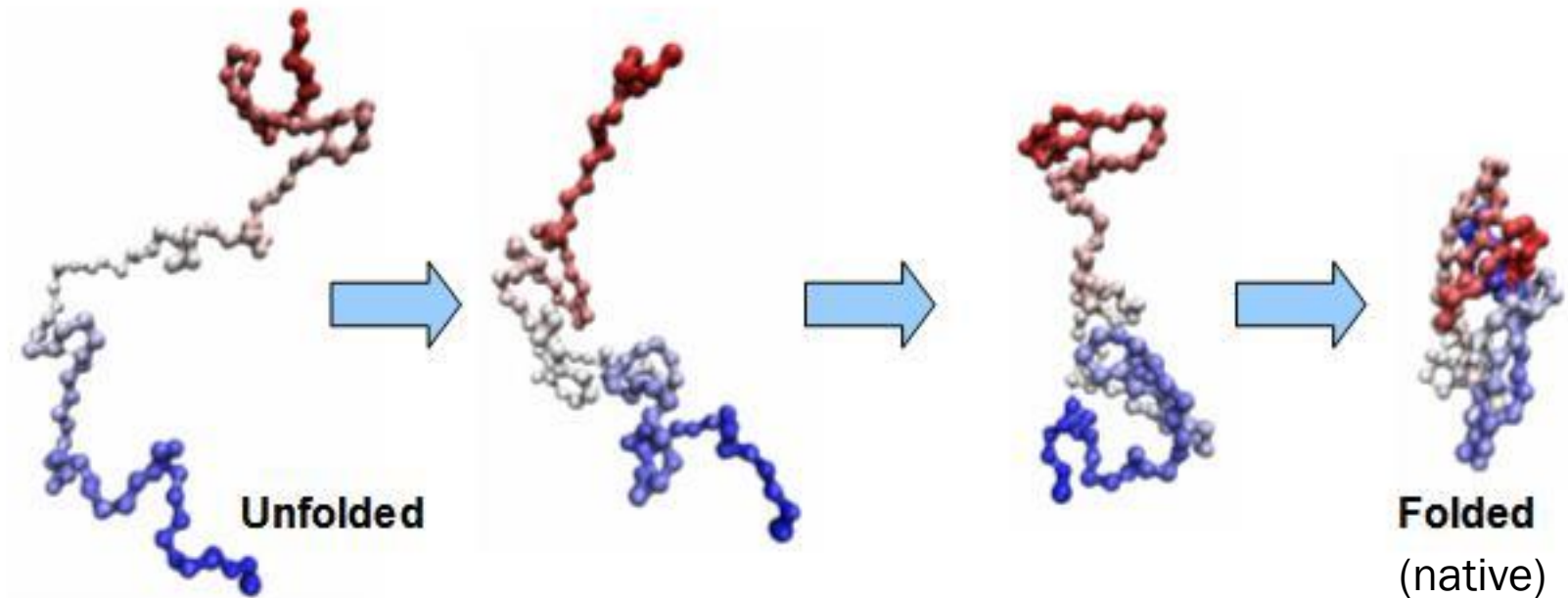


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

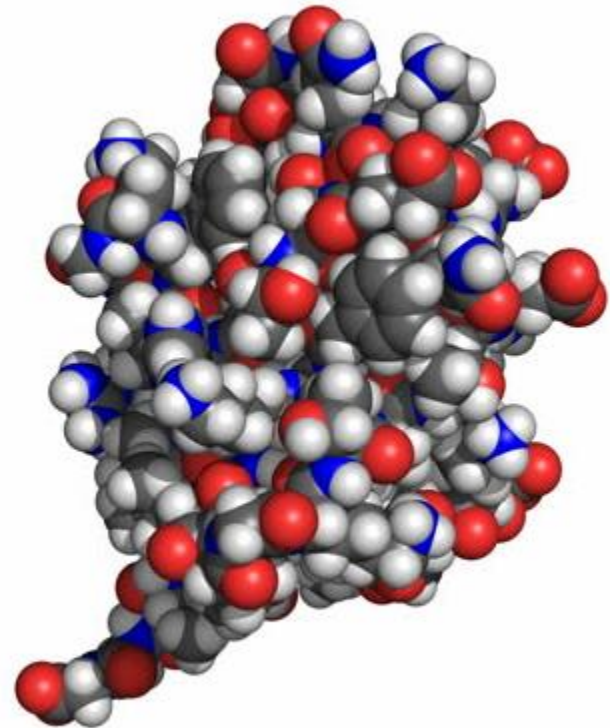
Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

Folding: How to Obtain 3D Structure

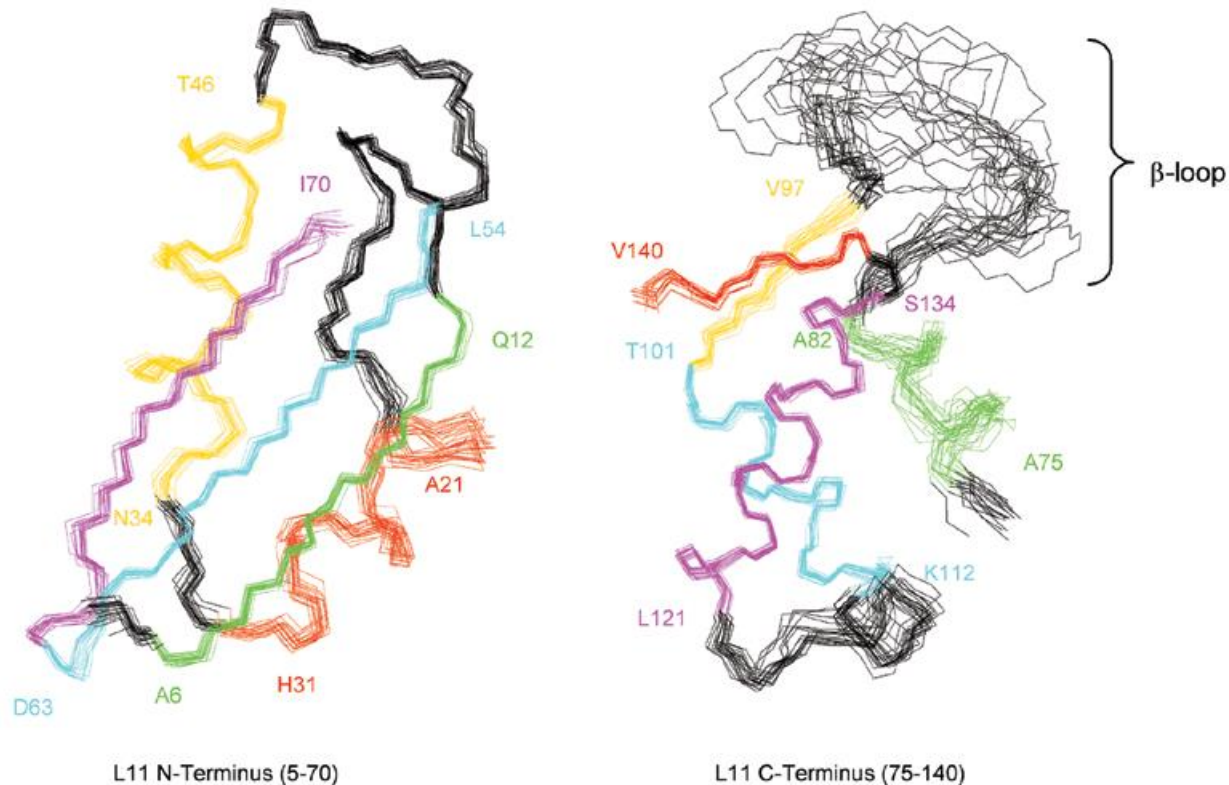


Folded Structure is Dense

- Packing density
 - Organic solid: 0.68 – 0.8
 - Proteins: 0.72 – 0.77



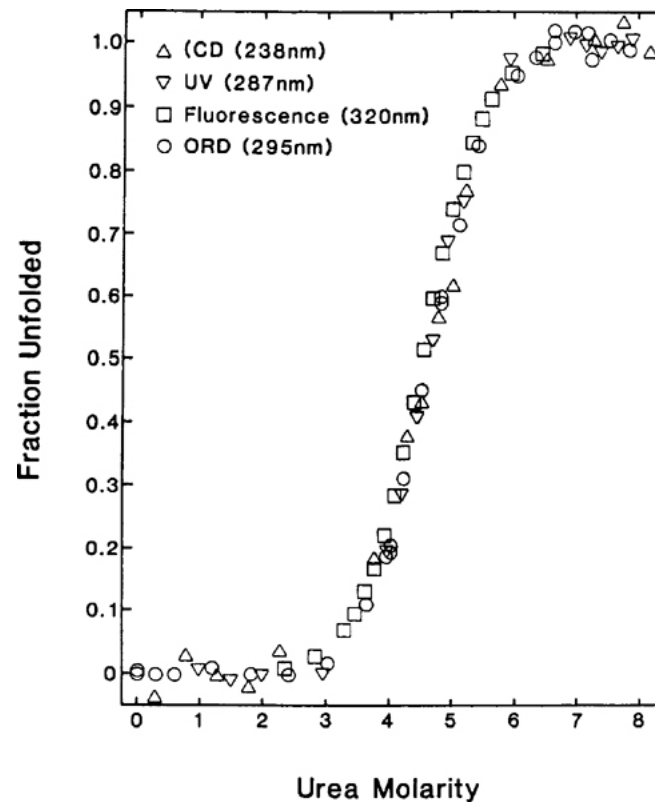
“Frozen” in the Folded State



(measurement by NMR)

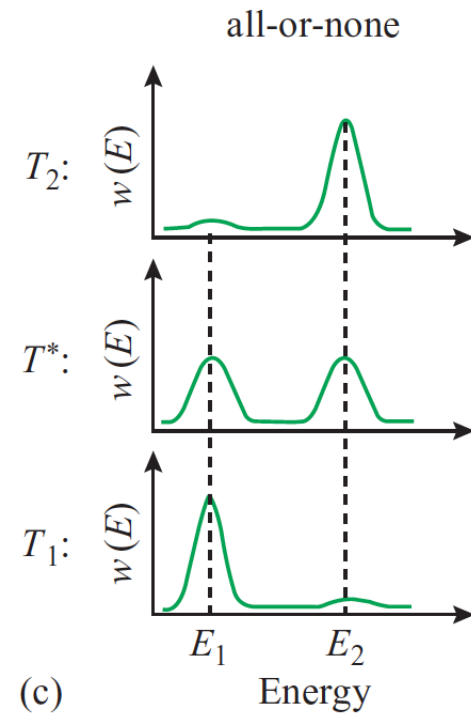
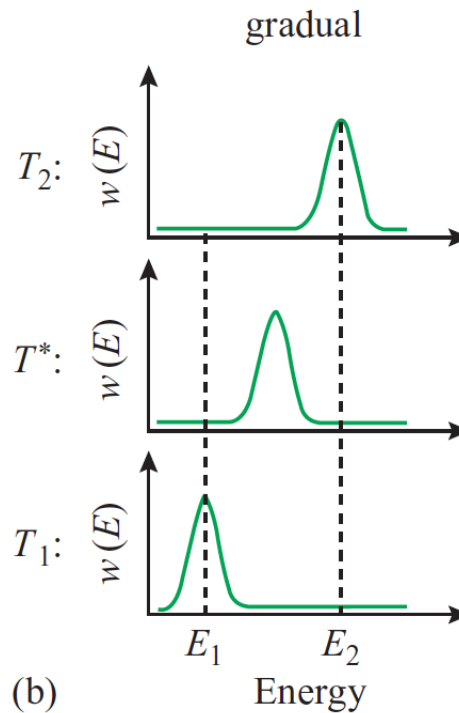
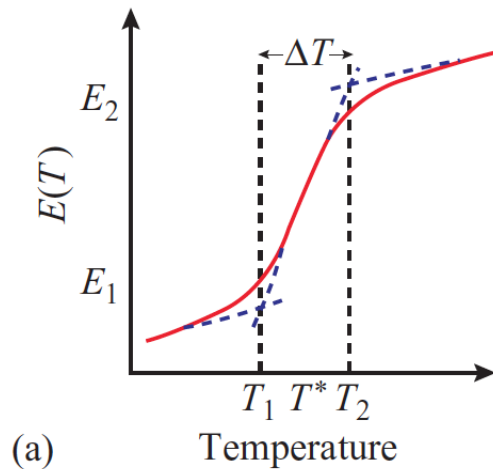
Image: <http://schwalbe.org.chemie.uni-frankfurt.de/node/684>

Protein Denaturation: S-shaped Curve



Shirley, "Urea and Guanidine Hydrochloride Denaturation Curves," in *Protein Stability and Folding* (1995)

Phase Transition?



van't Hoff Criterion

Assuming two-state folding,

$$P_{\text{fold}} = \frac{e^{-\beta G_{\text{folded}}}}{e^{-\beta G_{\text{folded}}} + e^{-\beta G_{\text{unfolded}}}} = \frac{1}{1 + e^{\beta \Delta G_{\text{fold}}}}$$

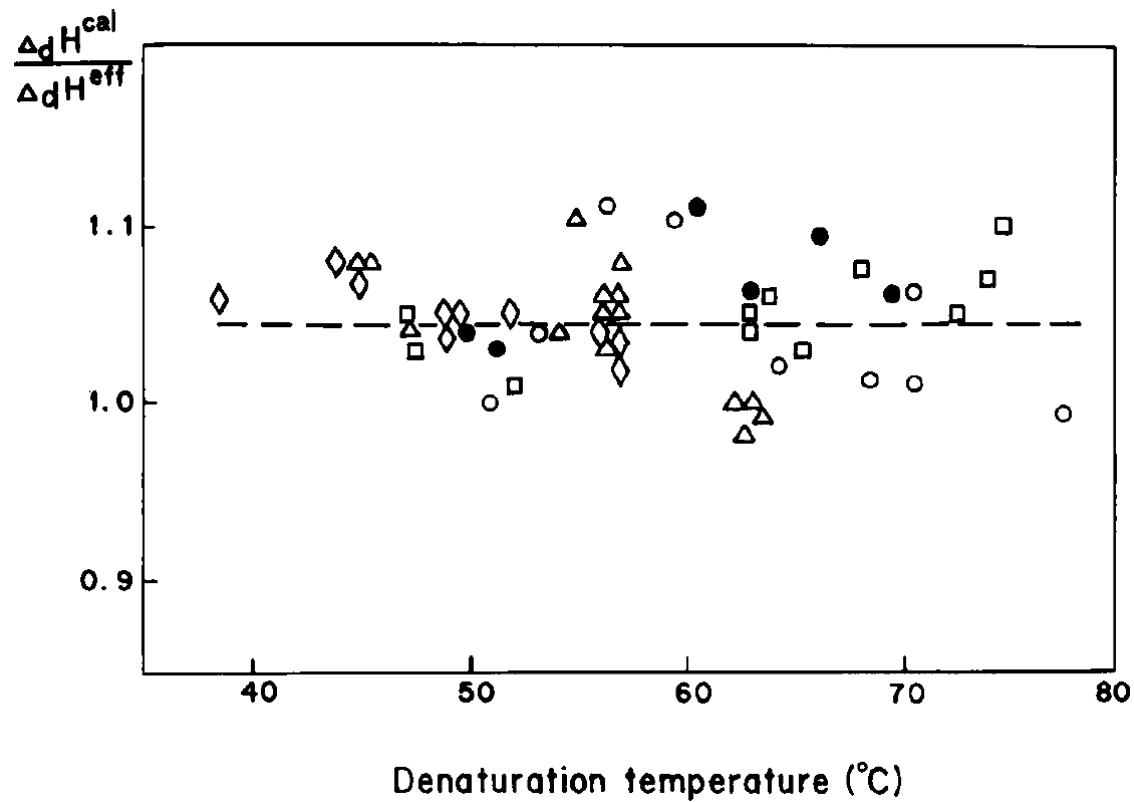
Using the Gibbs-Helmholtz equation,

$$\frac{dP_{\text{fold}}}{dT} = P_{\text{fold}}(1 - P_{\text{fold}}) \frac{\Delta H_{\text{fold}}}{k_{\text{B}} T^2}$$

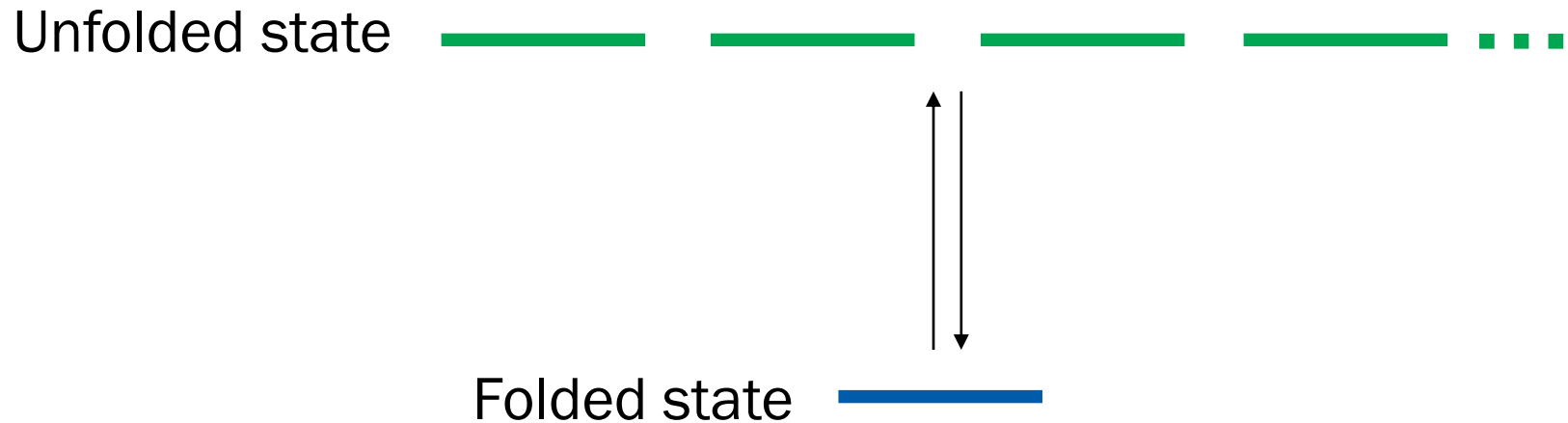
At the transition point ($P_{\text{fold}} = 1/2$),

$$\left. \frac{dP_{\text{fold}}}{dT} \right|_{T=T_f} = \frac{\Delta H_{\text{fold}}}{4k_{\text{B}} T_f^2}$$

Experimentally ...

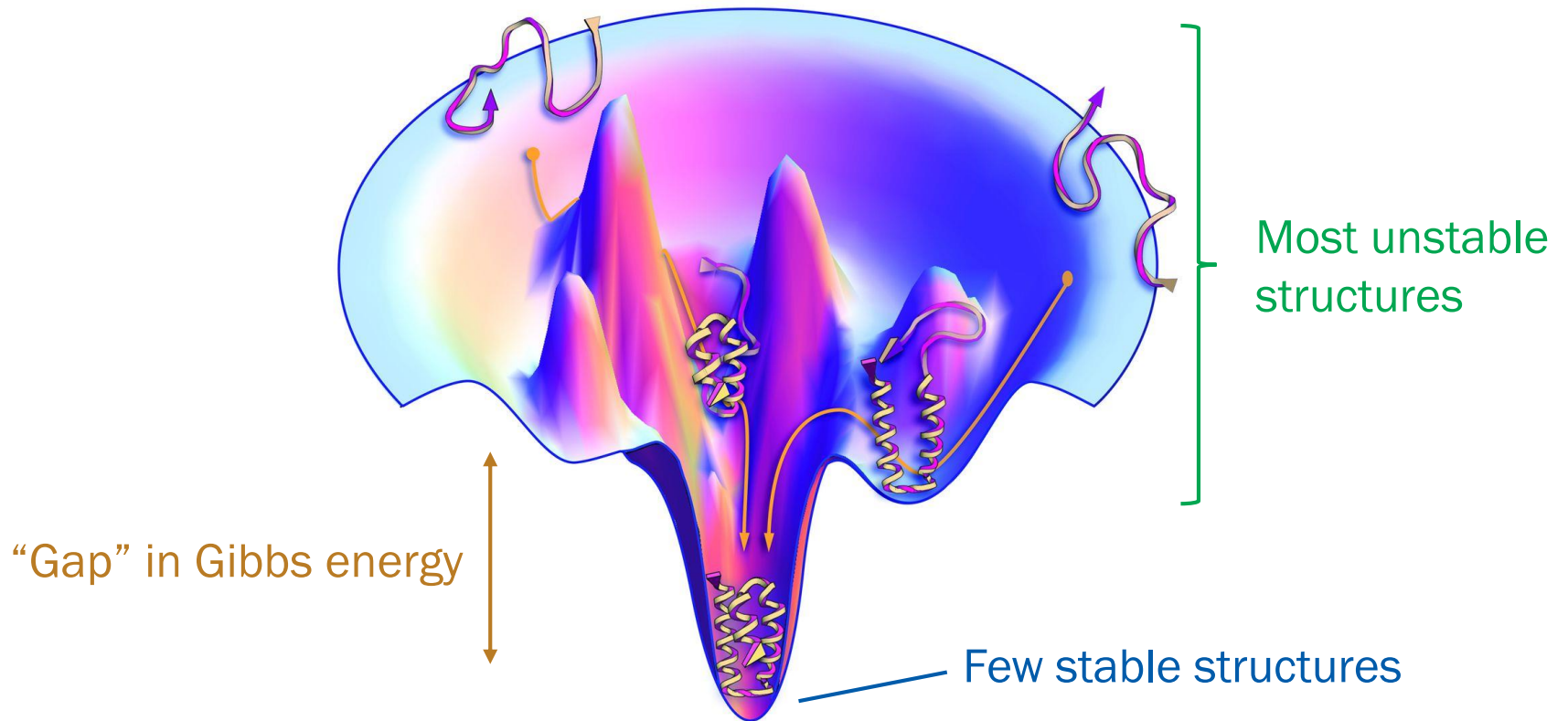


Two-State Model

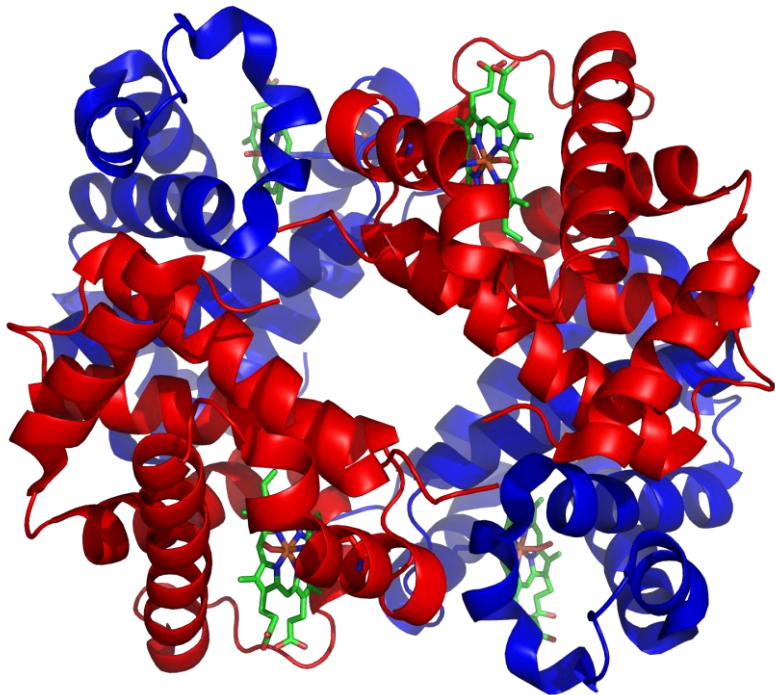


What is the underlying physics?

“Funnel” Picture

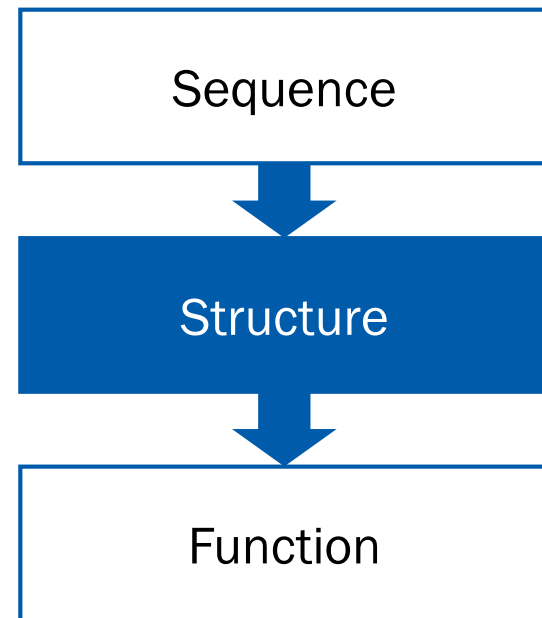


Why is Folding Important?

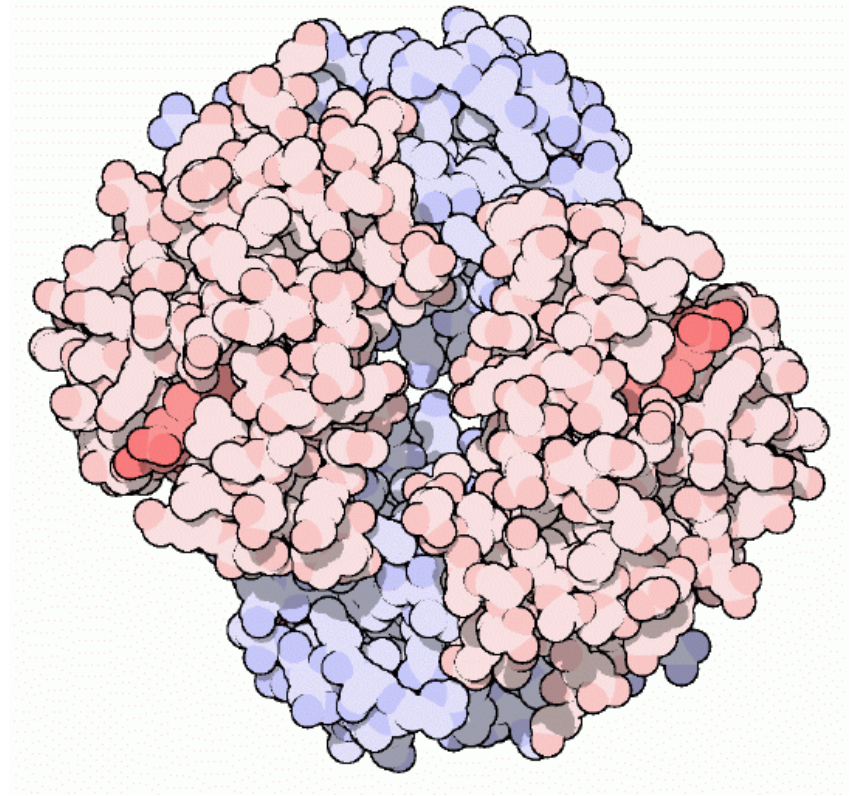
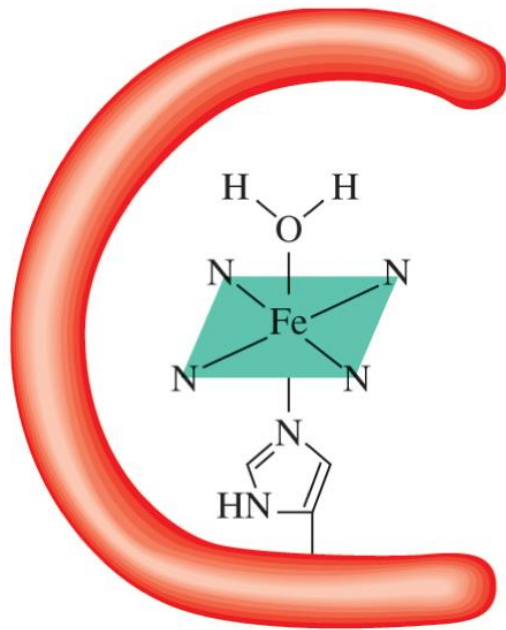


Hemoglobin

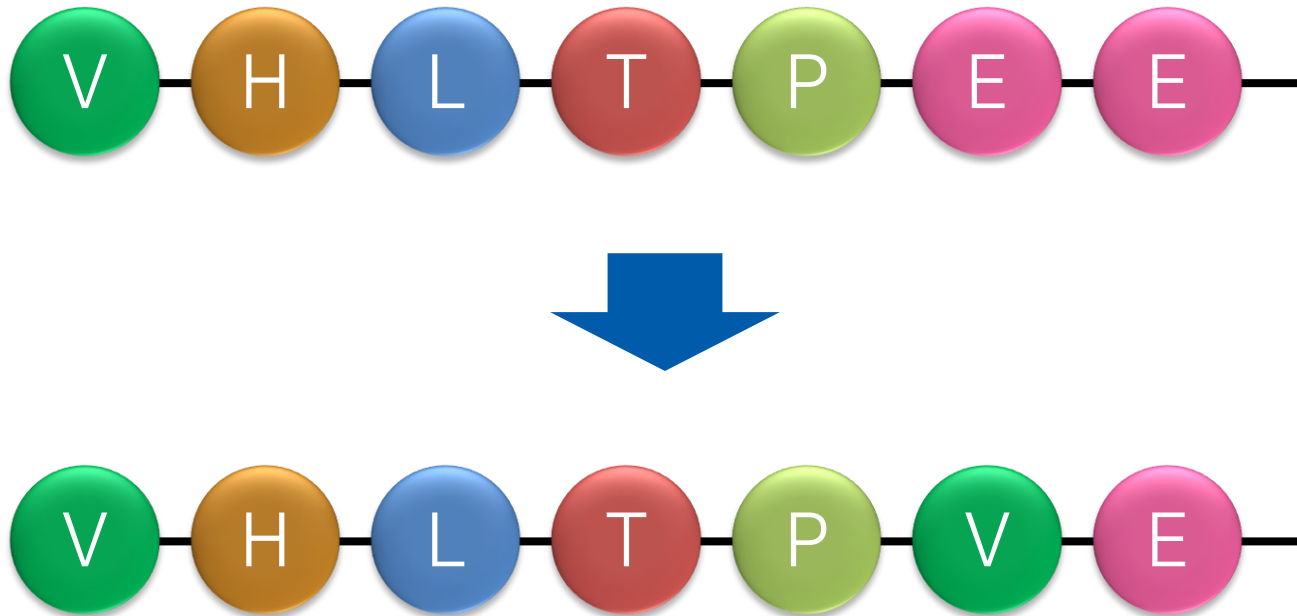
Structure-function paradigm



Structure-Function: Hemoglobin

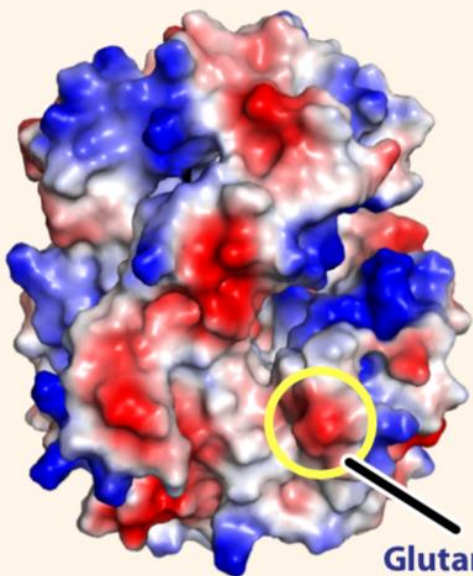
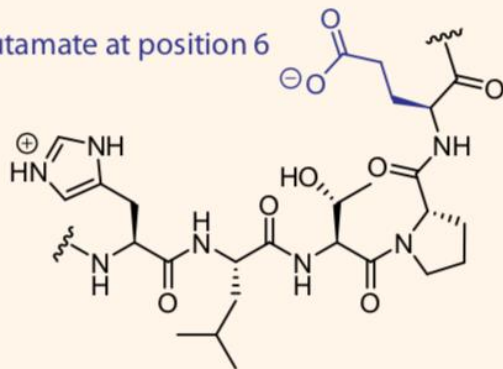


Sequence Change may Affect ...



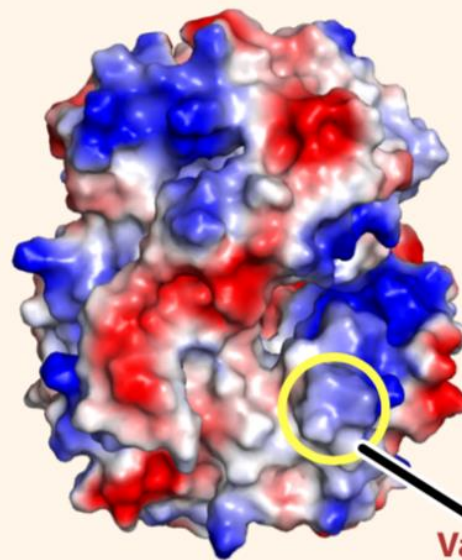
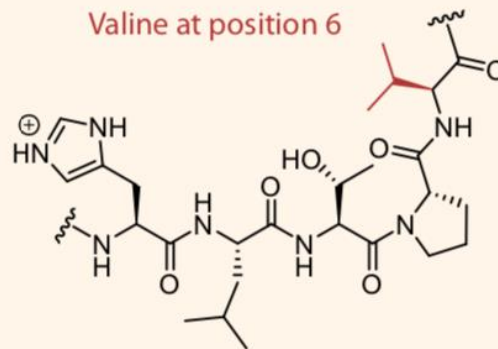
Surface Property

Glutamate at position 6



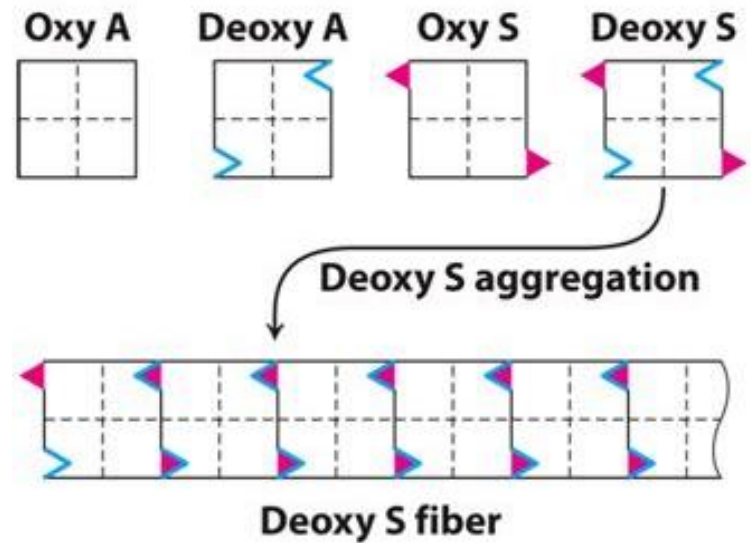
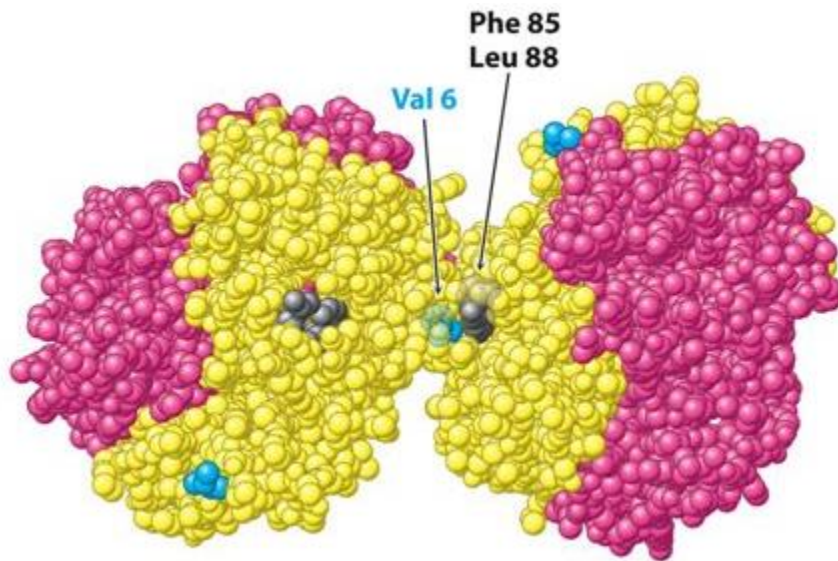
Glutamate

Valine at position 6



Valine

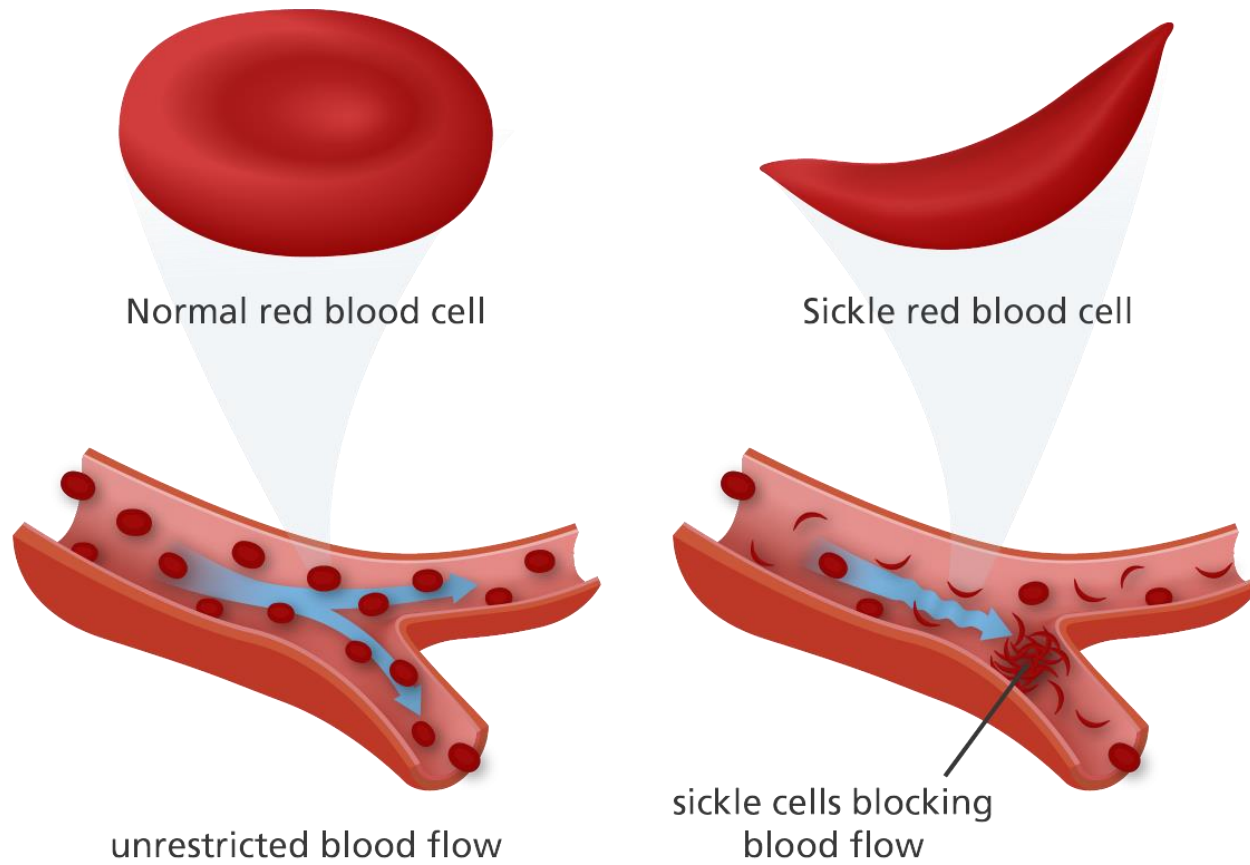
Molecular Interactions



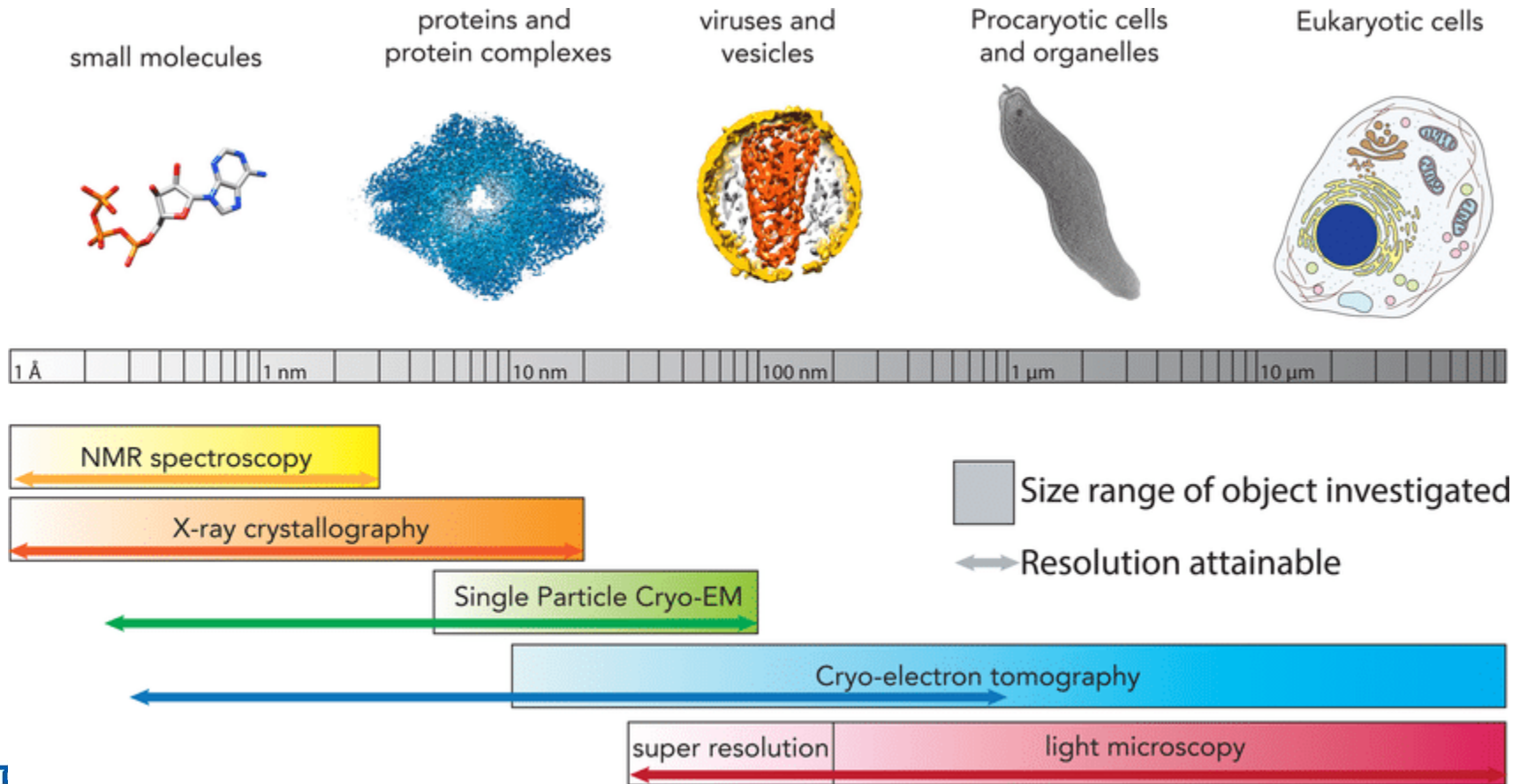
Cell Morphology



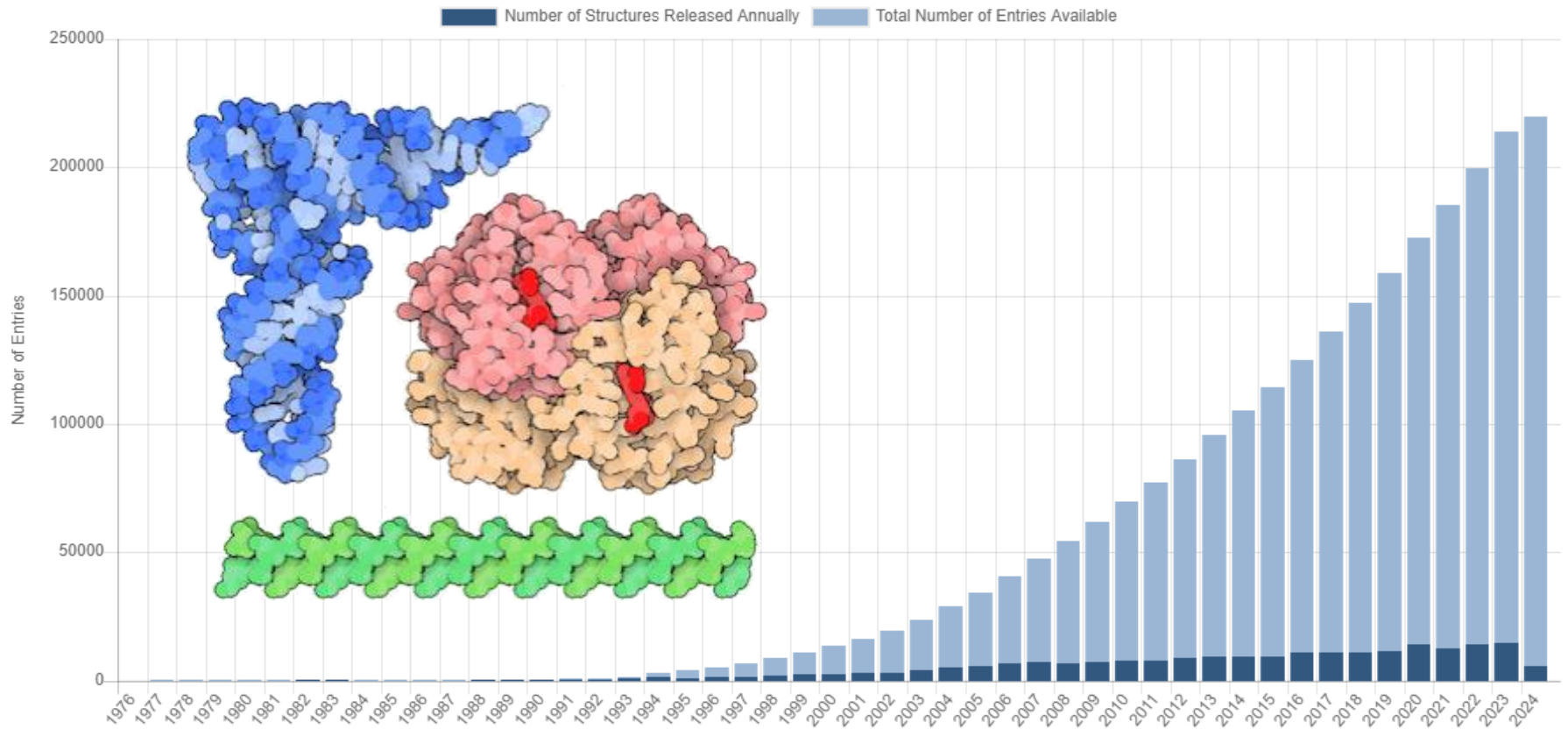
and even Pathology



Structural Biology

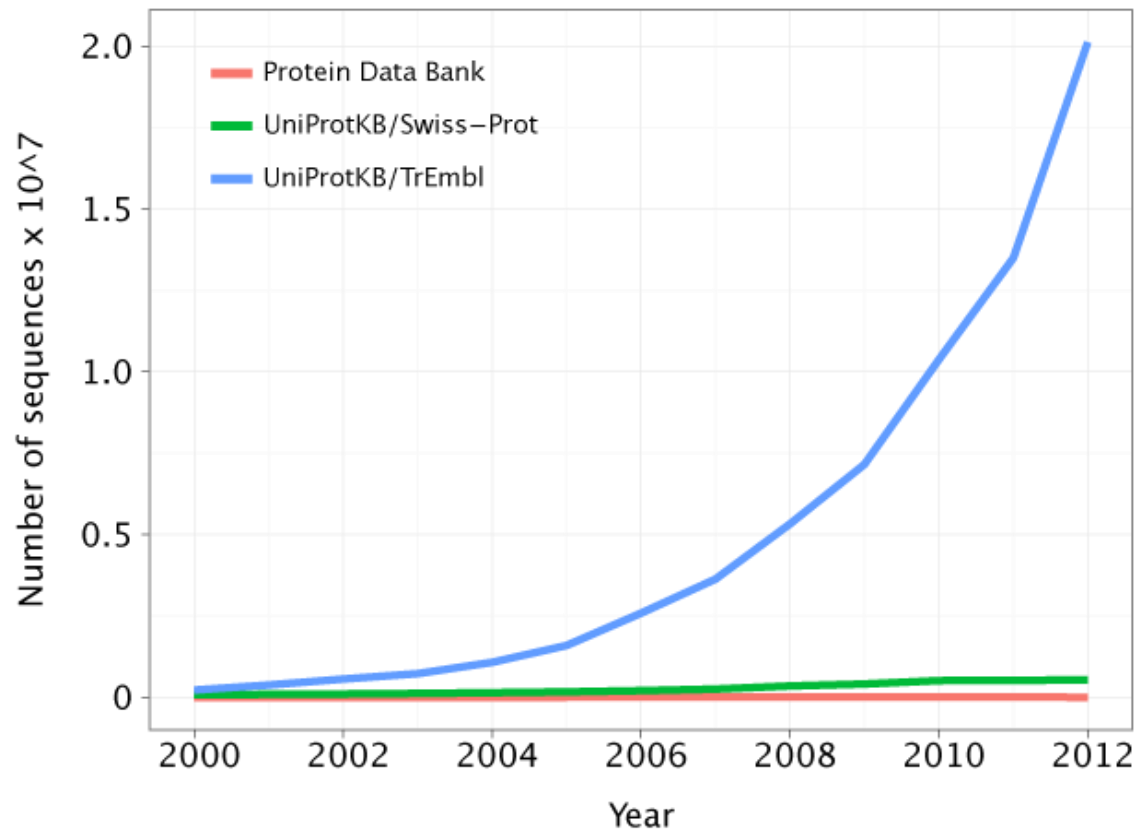


PDB: Structural Database

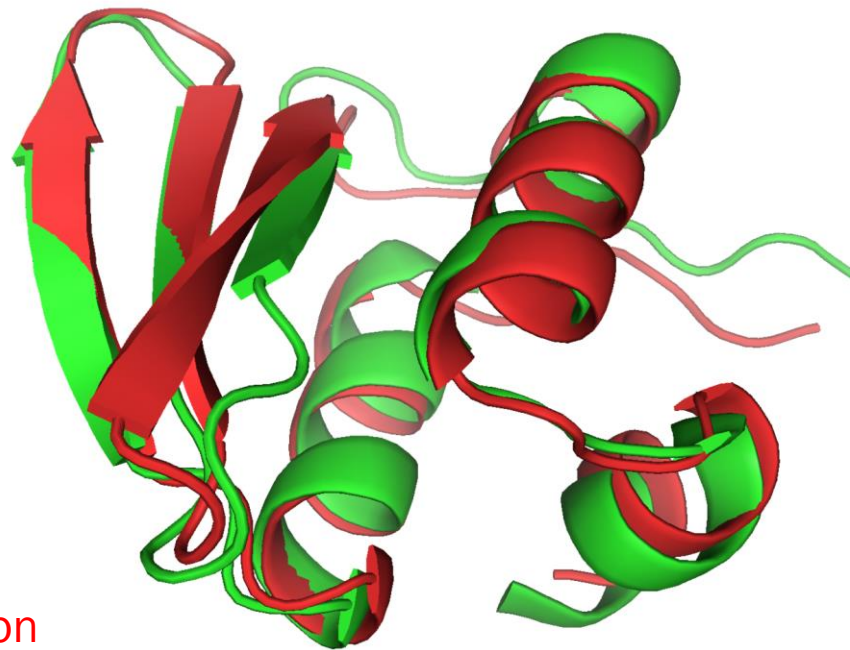


<https://www.rcsb.org/stats/growth/growth-released-structures>

Sequence Information is Cheaper



“Computational” Structural Biology

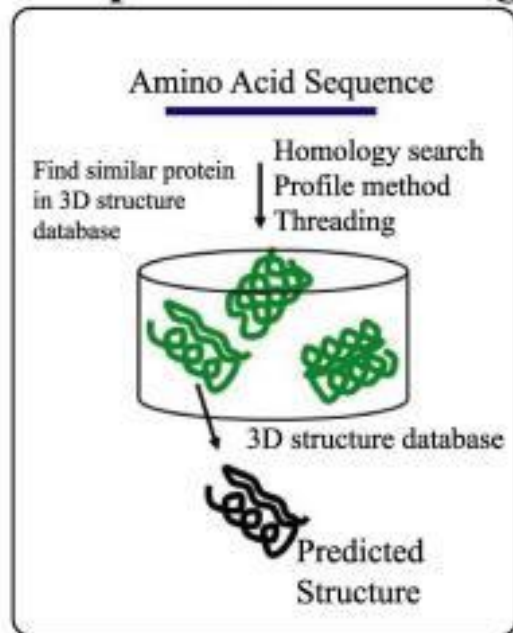


Red: prediction

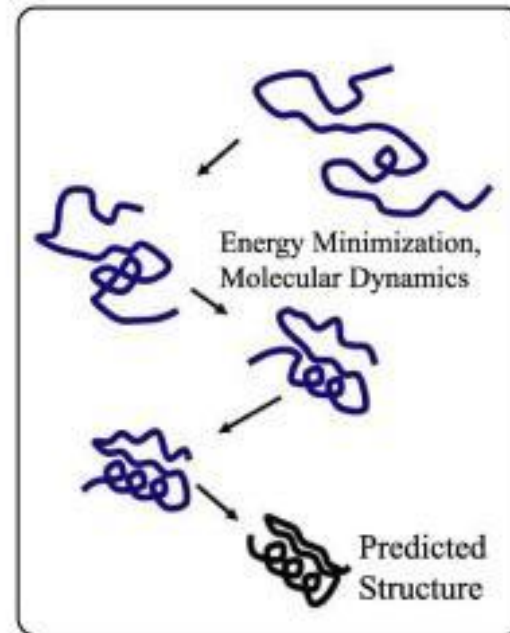
Green: experiment

How to Predict

Comparative Modelling

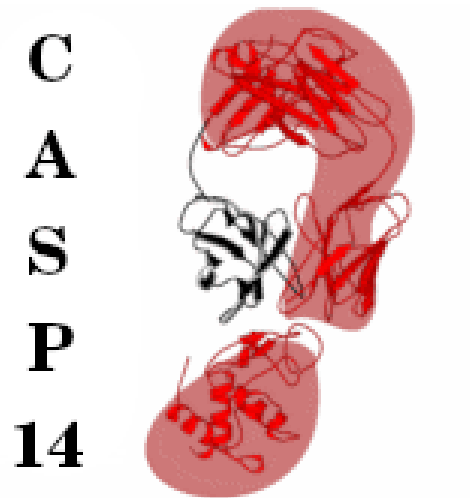


Ab initio Prediction

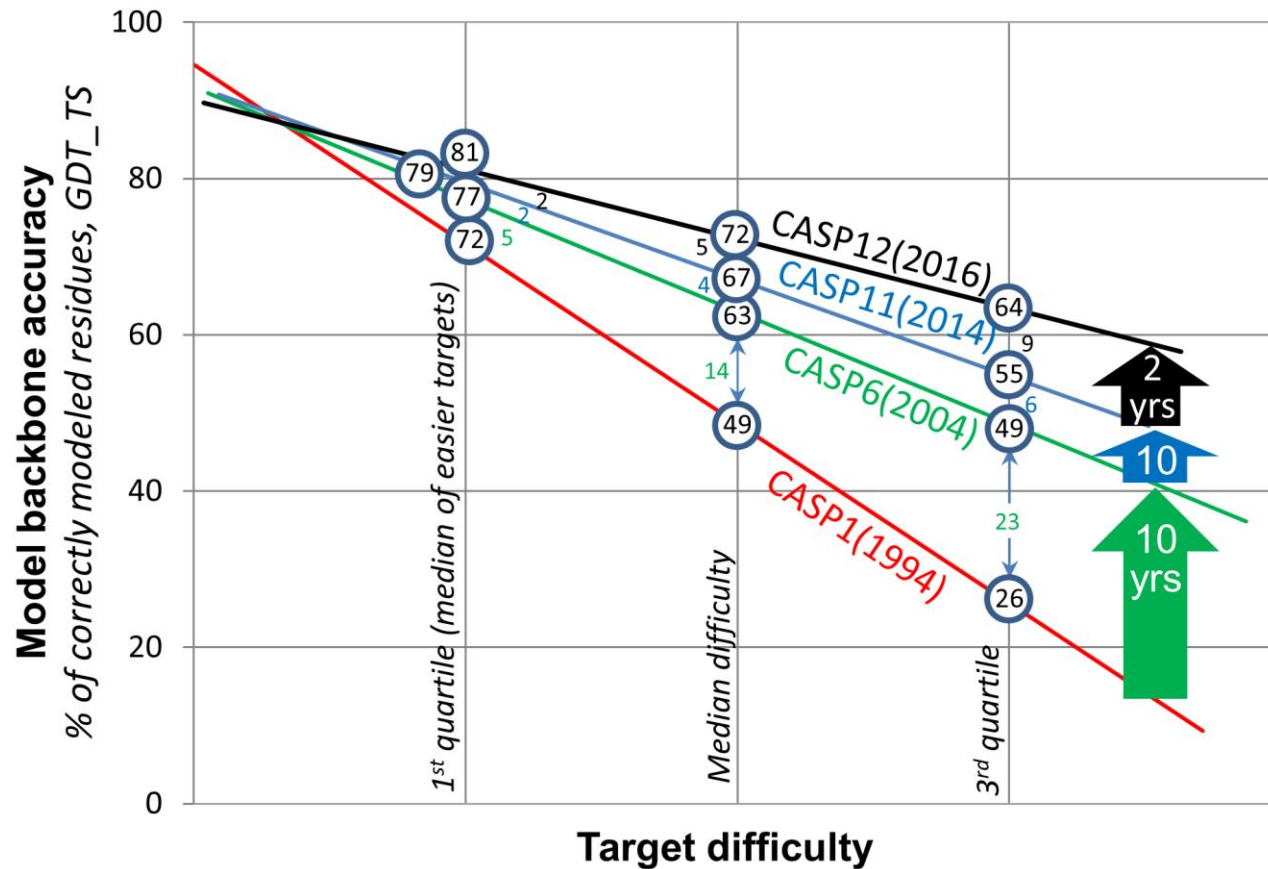


CASP

- Critical Assessment of protein Structure Prediction
(<http://predictioncenter.org/>)



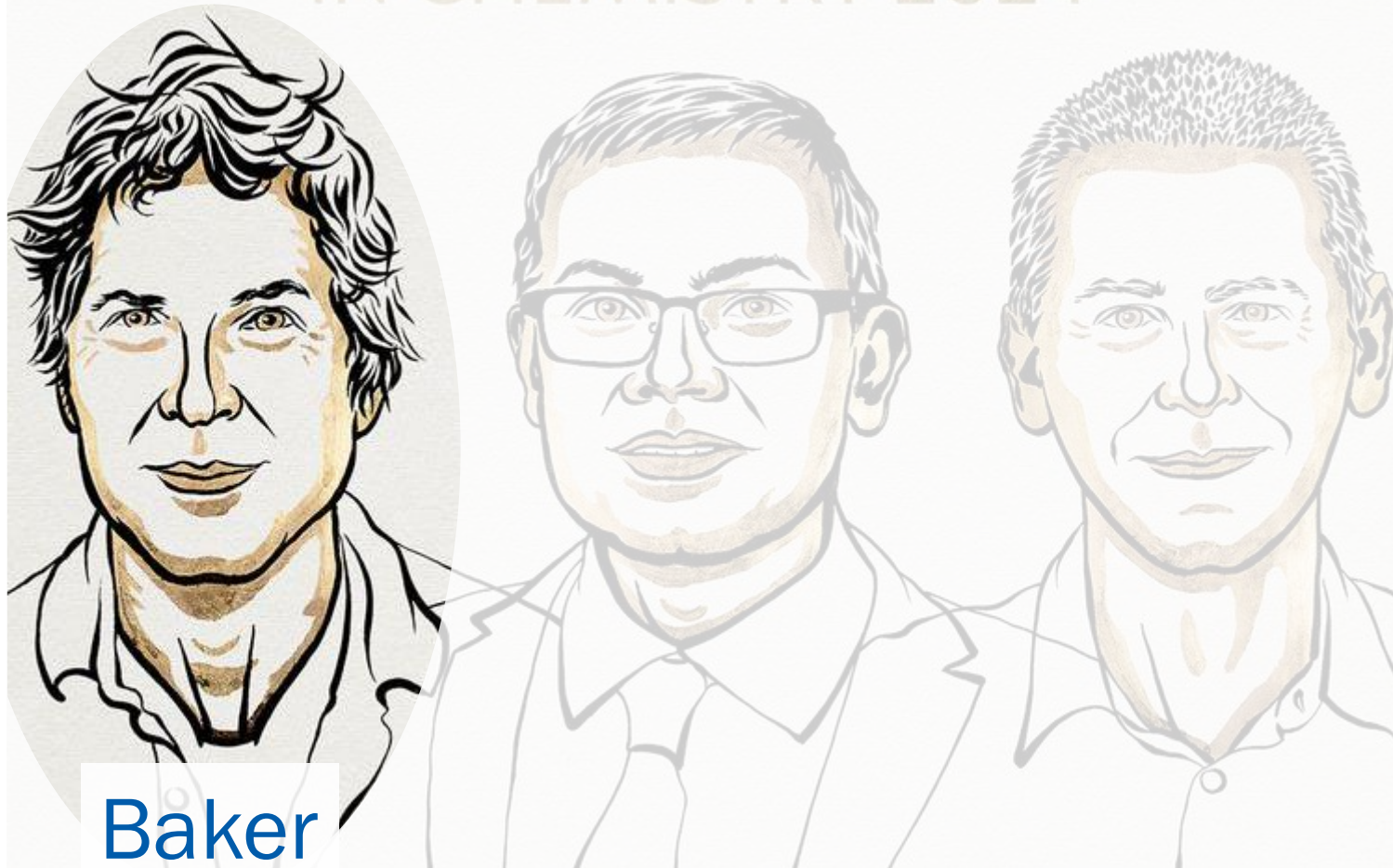
Accuracy of Prediction Models



THE NOBEL PRIZE IN CHEMISTRY 2024



Computational Protein Design



Baker

ROSETTA (1998)

PROTEINS: Structure, Function, and Genetics Suppl 3:171–176 (1999)

AB INITIO: PREDICTION REPORTS

Ab Initio Protein Structure Prediction of CASP III Targets Using ROSETTA

Kim T. Simons,¹ Rich Bonneau,¹ Ingo Ruczinski,² and David Baker^{1*}

¹*Department of Biochemistry, University of Washington, Seattle, Washington*

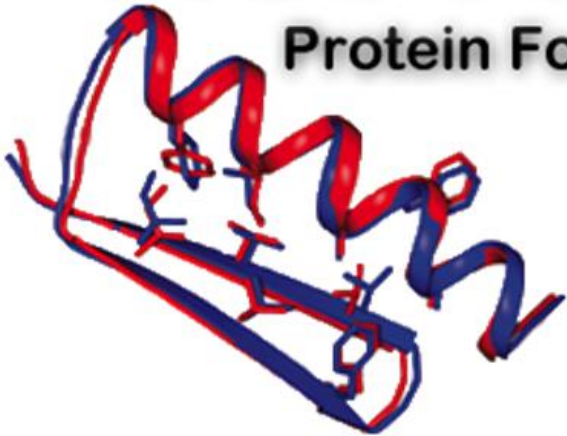
²*Department of Statistics, University of Washington, Seattle, Washington*



Rostta@home (2004)

Rosetta@home

Protein Folding, Design, and Docking



FoldIt (2008)

The screenshot displays the FoldIt game interface. In the center is a 3D model of a protein structure, colored in yellow, orange, and blue. To the left of the protein is a 'Recipe Output' window showing a list of commands and their results:

```
Recipe Output
Criter 7-rnd : -22.013
Criter 5-rnd : 20.297
Criter 8-rnd : 0.01
Gained another 1.509 pts.
Fuzzing...
Gained another 0.371 pts.
Gained another 3.029 pts.
Gained another 2.26 pts.
Criter 10-rnd : 9.474
Criter 6-rnd : -68.57
```

Below the recipe output is a 'Show script commands' button. To the left of the protein is a 'Pull Mode' button with a red 'X' and a 'Stop' button. Below these is a 'do_shake' button with 'Cancel' and 'Show Output' options. At the bottom left is a toolbar with icons for Shake, Mutate, Wiggle All, Wiggle Backbone, Wiggle Sidechains, Freeze Protein, Remove Bands, Disable Bands, Reset Structures, Reset Puzzle, Align Guide, Help, and Glossary. At the bottom center is a menu bar with options: Actions, Undo, Social, Modes, Behavior, View, and Menu. At the top right is a status bar showing Rank: 98, Score: 8919.576, and a timer for the Beginner Puzzle: Killer Toxin. At the bottom right is a competition leaderboard showing Group and Soloist competitions.

Rank: 98 Score: 8919.576
Soloist Beginner Puzzle: Killer Toxin
Expires 1/08/2013 0:00 MZ (29 days, 9 hours)
► No bonuses or conditions

Group Competition

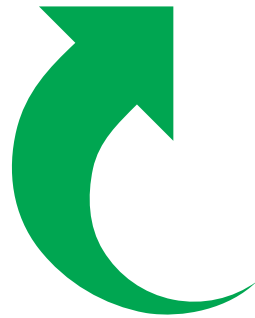
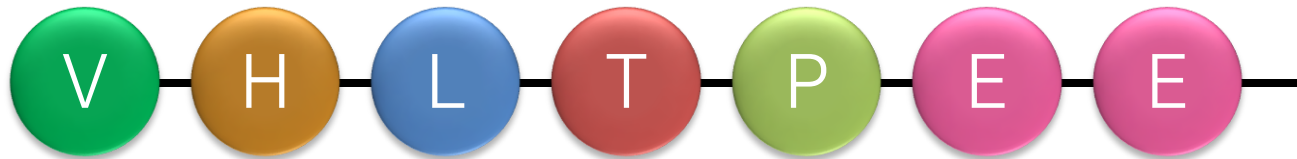
#	Group Name	Score
---	------------	-------

Soloist Competition

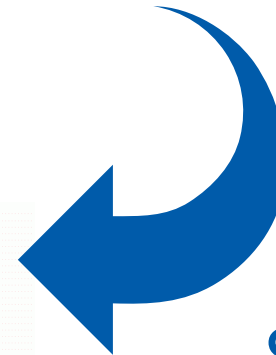
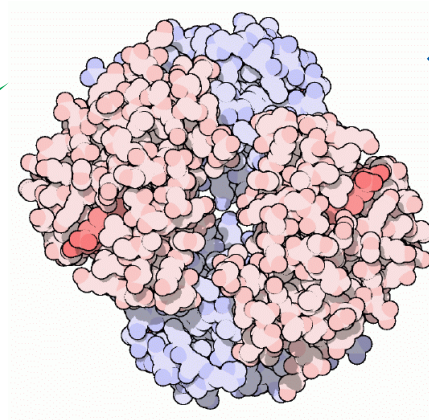
#	Player Name	Current	Best
1	mirp	9916	9982
2	bertro	-	9972
3	pauldunn	-	9920
4	O Seki To	9873	9909
5	ghwut	-	9904
6	hansvandenhof	-	9904
7	MooMooMan	9891	9902

Chat - Puzzle auto show
Chat - Global auto show
Notifications auto show

Protein Design?

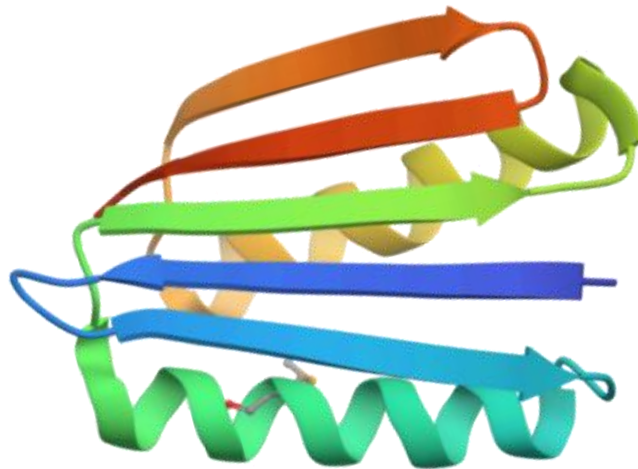


Protein design



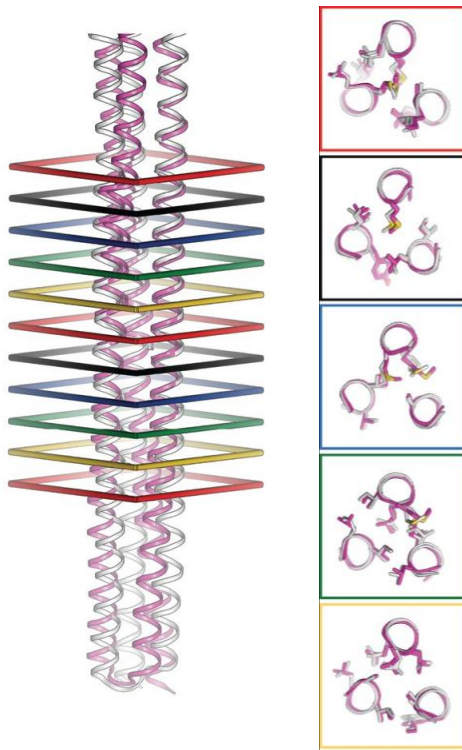
Structure
prediction

Top7 (2003)

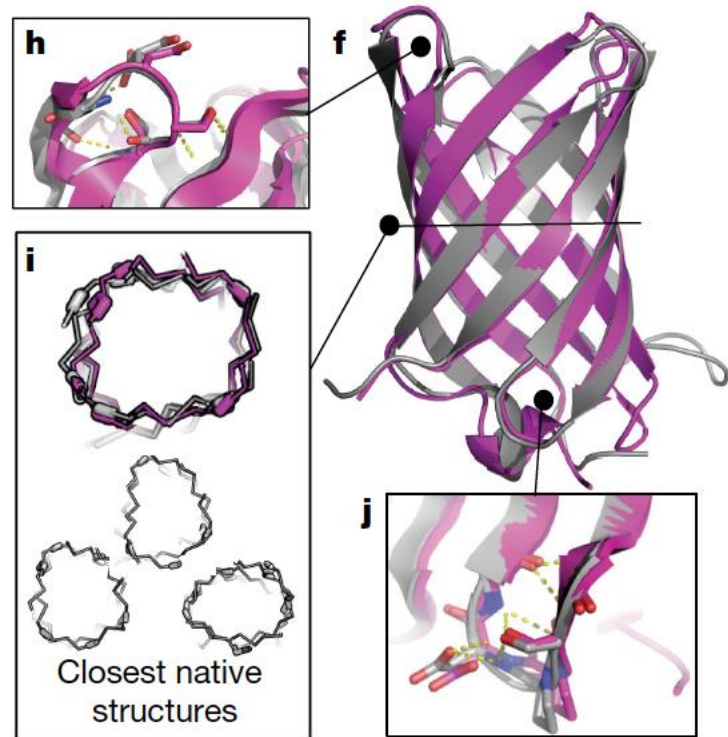


De novo Structure Design

α structure

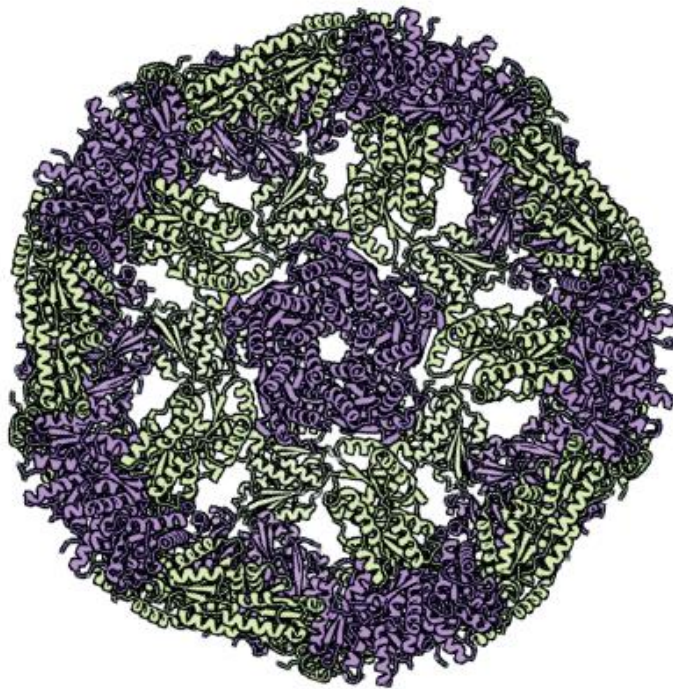


β structure

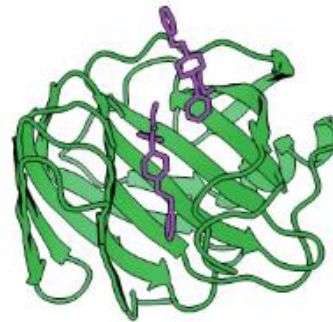


Huang et al., *Science* (2014); Dou et al., *Nature* (2018)

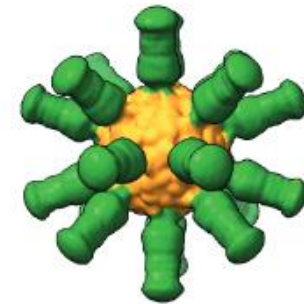
Recent Examples



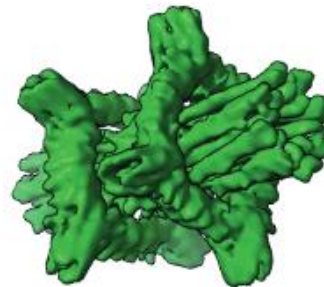
2016: New nanomaterials where up to 120 proteins spontaneously link together.



2017: Proteins that bind to an opioid called fentanyl (purple). These could be used to detect fentanyl in the environment.



2021: Nanoparticles (yellow) with proteins imitating influenza virus on the surface (green) that can be used as a vaccine for influenza. Successful in animal models.

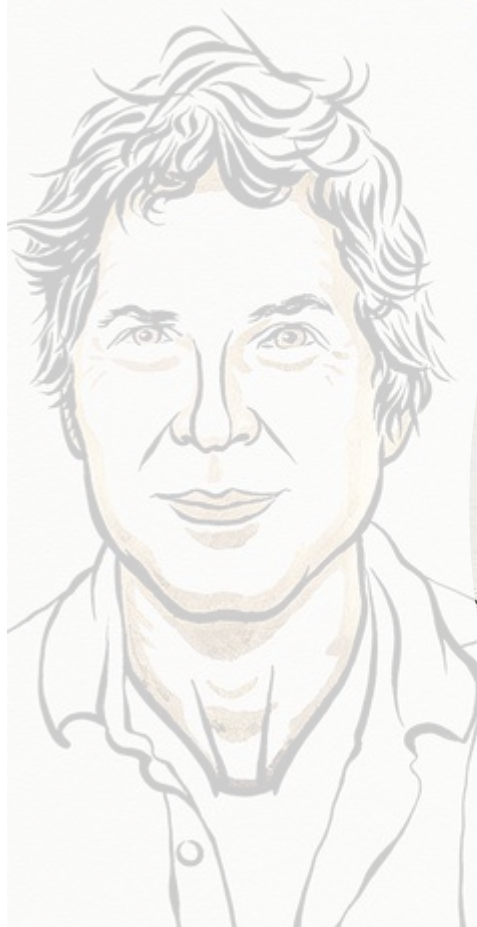


2022: Proteins that function as a type of molecular rotor.



2024: Geometrically shaped proteins that can change their shape due to external influences. Could be used for producing tiny sensors.

Protein Structure Prediction



Hassabis

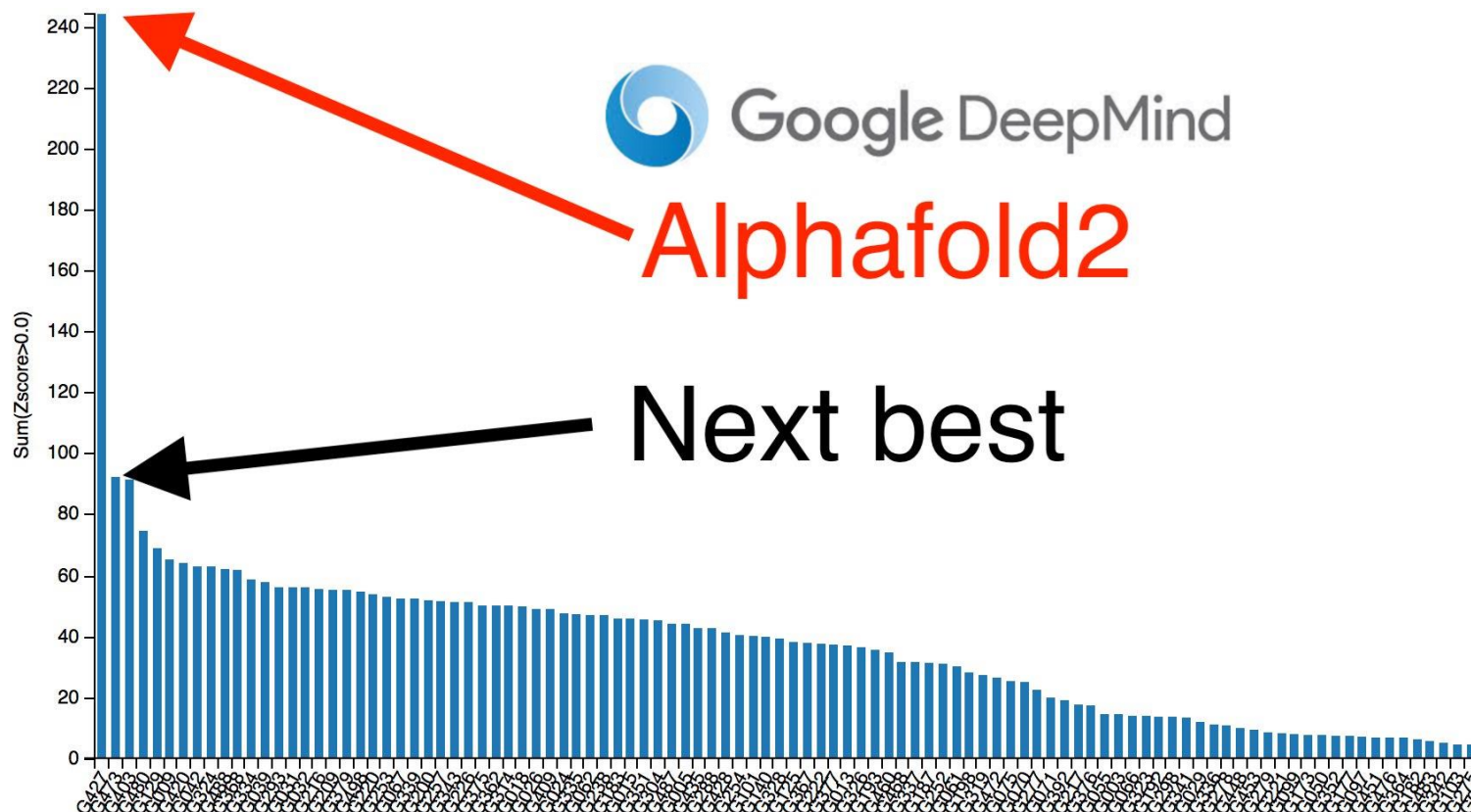


Jumper

AlphaGo (2016)



Shocking News from CASP14 (2020)



AlphaFold 2

Article

Highly accurate protein structure prediction with AlphaFold

<https://doi.org/10.1038/s41586-021-03819-2>

Received: 11 May 2021

Accepted: 12 July 2021

Published online: 15 July 2021

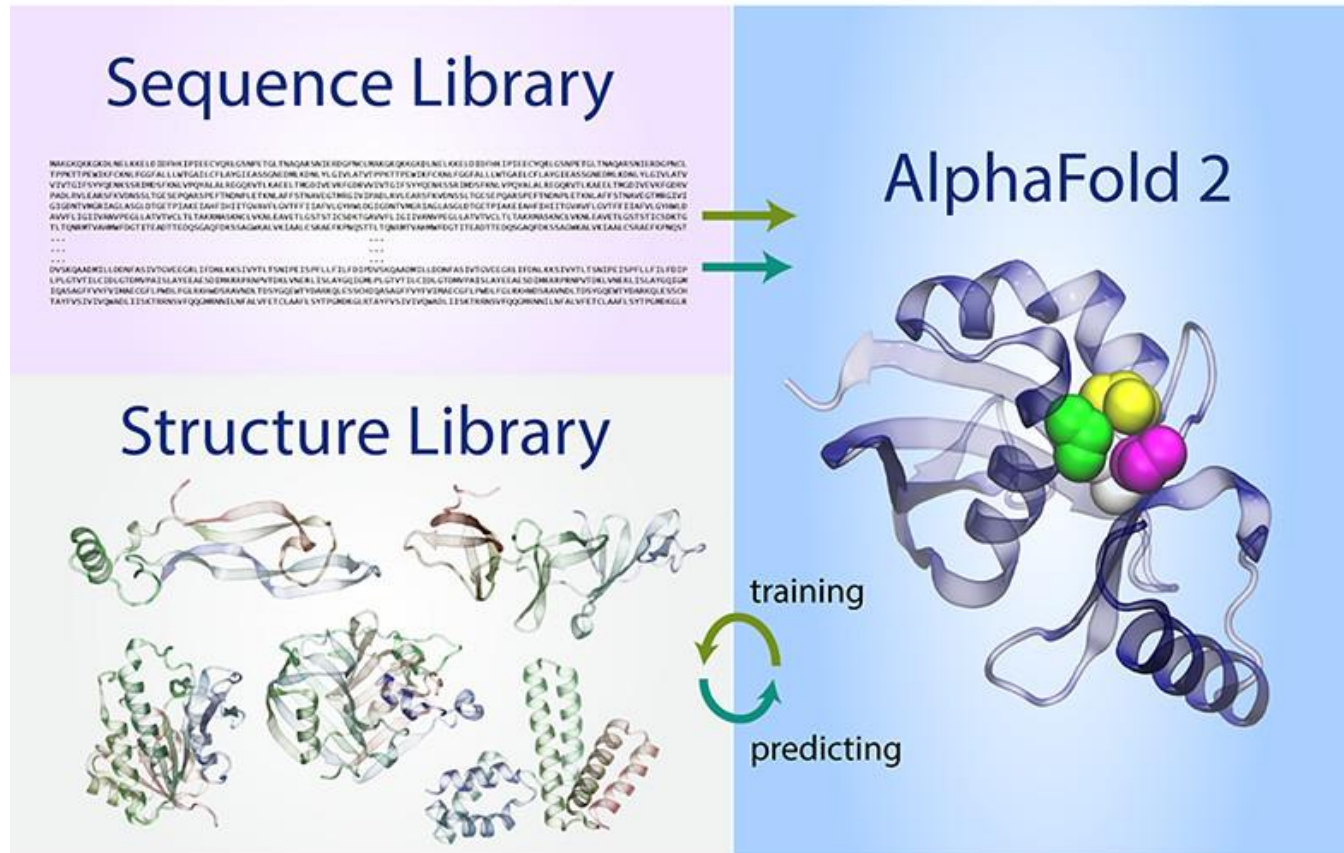
Open access



Check for updates

John Jumper^{1,4}✉, Richard Evans^{1,4}, Alexander Pritzel^{1,4}, Tim Green^{1,4}, Michael Figurnov^{1,4}, Olaf Ronneberger^{1,4}, Kathryn Tunyasuvunakool^{1,4}, Russ Bates^{1,4}, Augustin Žídek^{1,4}, Anna Potapenko^{1,4}, Alex Bridgland^{1,4}, Clemens Meyer^{1,4}, Simon A. A. Kohl^{1,4}, Andrew J. Ballard^{1,4}, Andrew Cowie^{1,4}, Bernardino Romera-Paredes^{1,4}, Stanislav Nikolov^{1,4}, Rishub Jain^{1,4}, Jonas Adler¹, Trevor Back¹, Stig Petersen¹, David Reiman¹, Ellen Clancy¹, Michal Zielinski¹, Martin Steinegger^{2,3}, Michalina Pacholska¹, Tamas Berghammer¹, Sebastian Bodenstein¹, David Silver¹, Oriol Vinyals¹, Andrew W. Senior¹, Koray Kavukcuoglu¹, Pushmeet Kohli¹ & Demis Hassabis^{1,4}✉

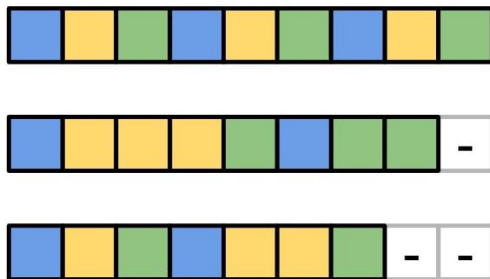
How It Works



Sequence Information

MS

(Multiple sequences)



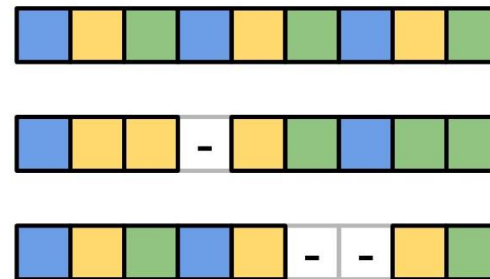
Align



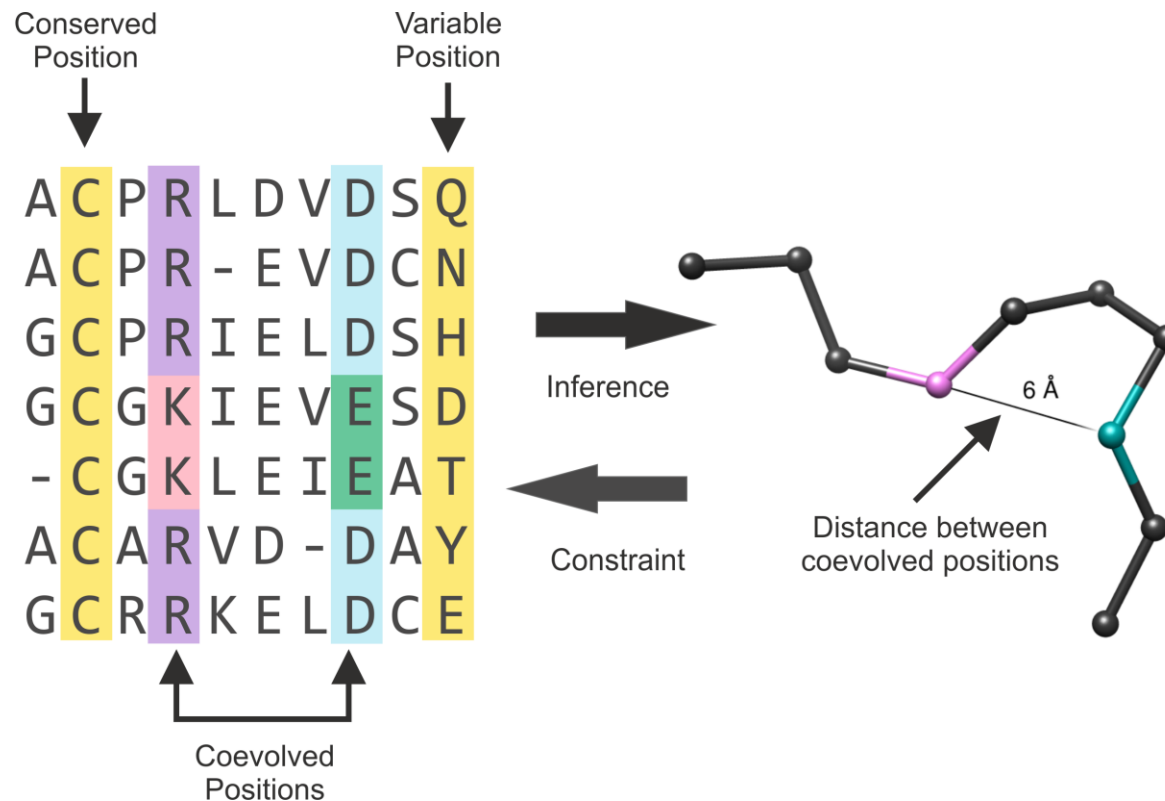
hhblits, blast,
jackhmmer,
clustal, etc.

MSA

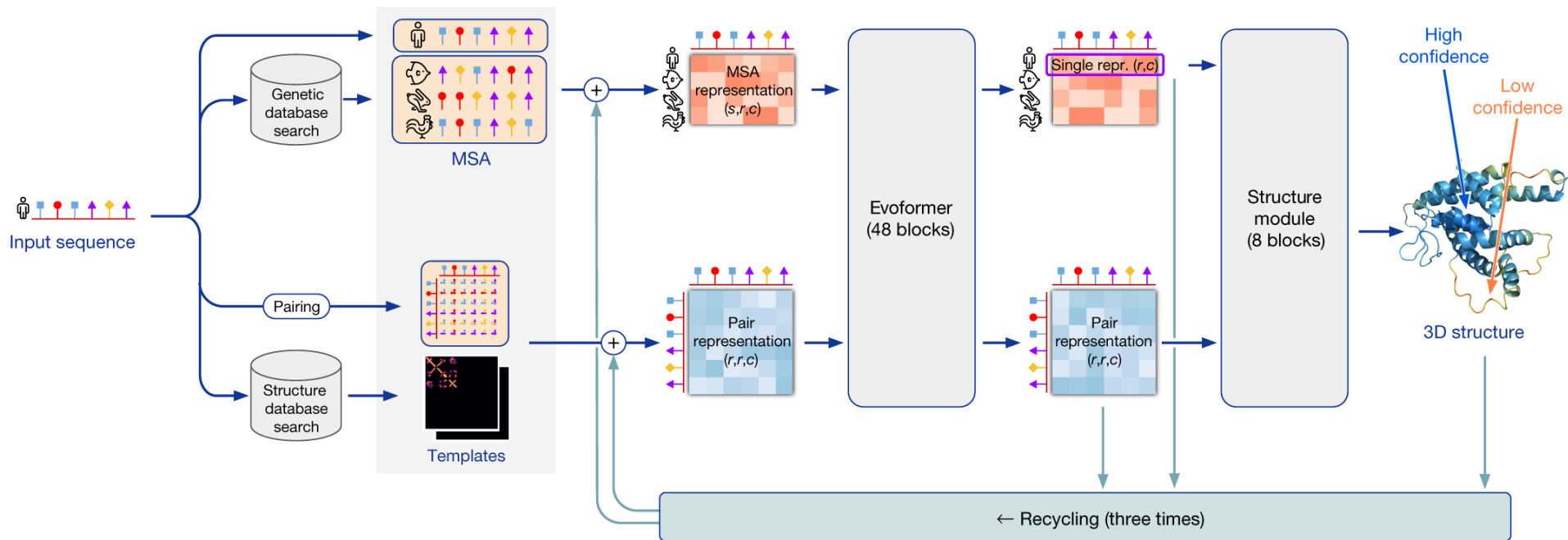
(Multiple sequence alignment)



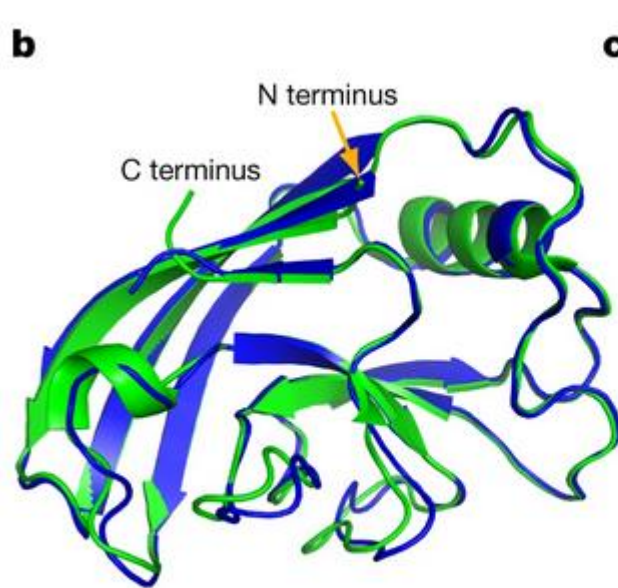
MSA to 3D Information



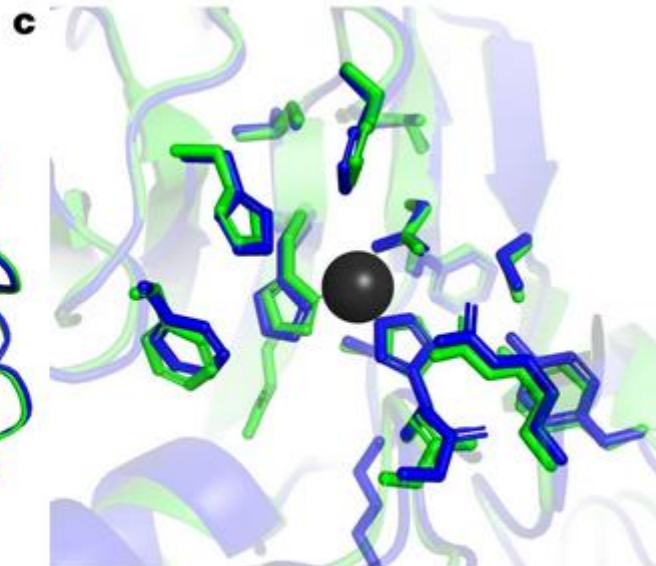
Secret Sauce: Deep Learning



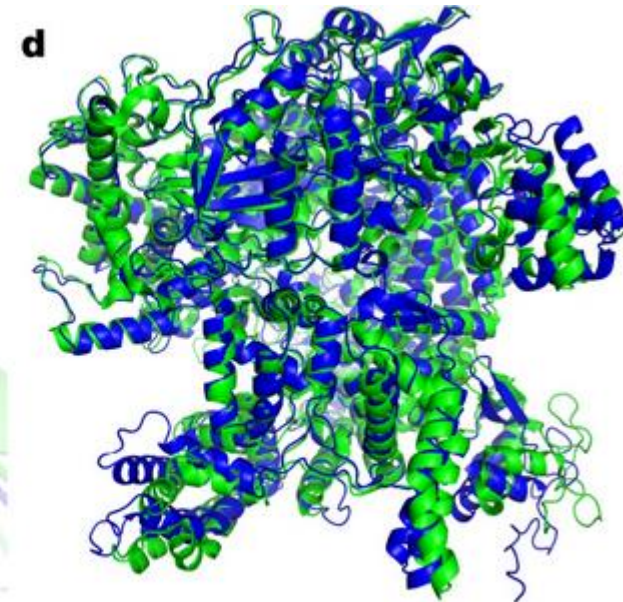
Unbelievable Accuracy



AlphaFold Experiment
r.m.s.d.₉₅ = 0.8 Å; TM-score = 0.93

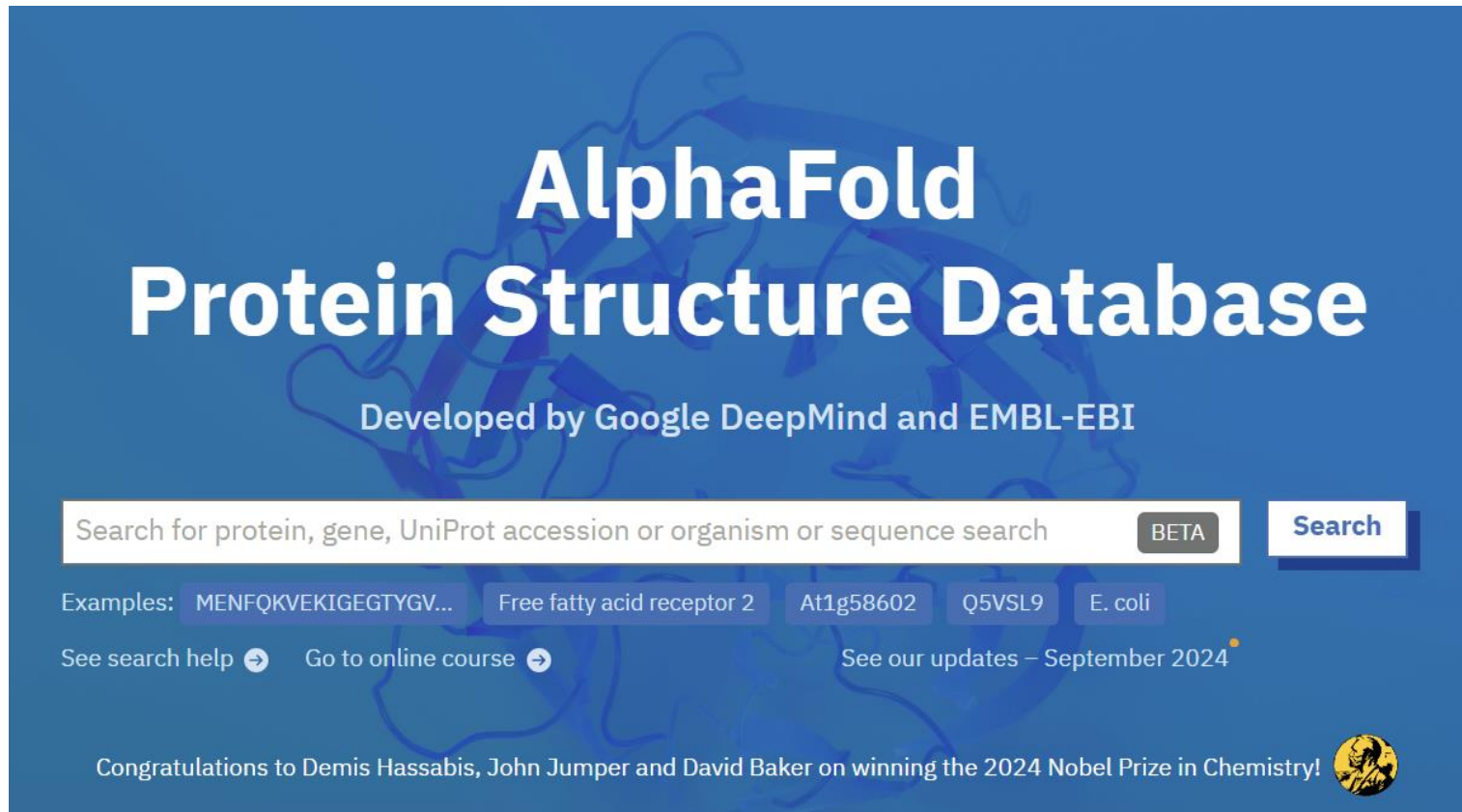


AlphaFold Experiment
r.m.s.d. = 0.59 Å within 8 Å of Zn



AlphaFold Experiment
r.m.s.d.₉₅ = 2.2 Å; TM-score = 0.96

200 M+ Predicted Structures



AlphaFold
Protein Structure Database

Developed by Google DeepMind and EMBL-EBI

Search for protein, gene, UniProt accession or organism or sequence search BETA Search

Examples: MENFQKVEKIGEGTYGV... Free fatty acid receptor 2 At1g58602 Q5VSL9 E. coli

[See search help](#) [Go to online course](#) [See our updates – September 2024](#)

Congratulations to Demis Hassabis, John Jumper and David Baker on winning the 2024 Nobel Prize in Chemistry! 

After AlphaFold 2

Article

Highly accurate protein structure prediction with AlphaFold

<https://doi.org/10.1038/s41586-021-03819-2>

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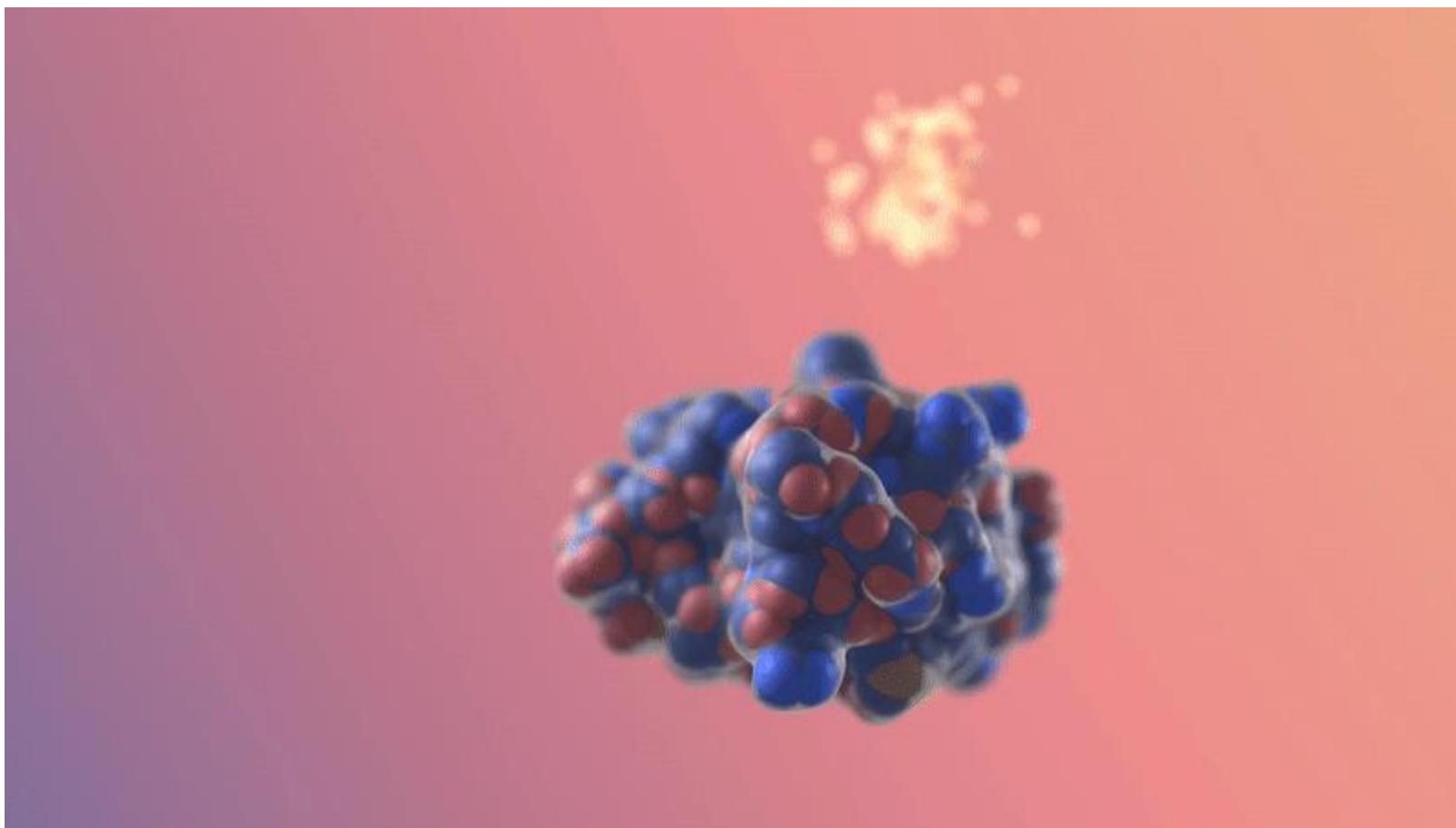
Check for updates

John Jumper^{1,4}✉, Richard Evans^{1,4}, Alexander Pritzel^{1,4}, Tim Green^{1,4}, Michael Figurnov^{1,4}, Olaf Ronneberger^{1,4}, Kathryn Tunyasuvunakool^{1,4}, Russ Bates^{1,4}, Augustin Žídek^{1,4}, Anna Potapenko^{1,4}, Alex Bridgland^{1,4}, Clemens Meyer^{1,4}, Simon A. A. Kohl^{1,4}, Andrew J. Ballard^{1,4}, Andrew Cowie^{1,4}, Bernardino Romera-Paredes^{1,4}, Stanislav Nikolov^{1,4}, Rishub Jain^{1,4}, Jonas Adler¹, Trevor Back¹, Stig Petersen¹, David Reiman¹, Ellen Clancy¹, Michal Zielinski¹, Martin Steinegger^{2,3}, Michalina Pacholska¹, Tamas Berghammer¹, Sebastian Bodenstein¹, David Silver¹, Oriol Vinyals¹, Andrew W. Senior¹, Koray Kavukcuoglu¹, Pushmeet Kohli¹ & Demis Hassabis^{1,4}✉

RosettaFold (2021)



RFDiffusion (2023)



AlphaFold 3 (2024)

Article

Accurate structure prediction of biomolecular interactions with AlphaFold 3

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 Check for updates

Josh Abramson^{1,7}, Jonas Adler^{1,7}, Jack Dunger^{1,7}, Richard Evans^{1,7}, Tim Green^{1,7}, Alexander Pritzel^{1,7}, Olaf Ronneberger^{1,7}, Lindsay Willmore^{1,7}, Andrew J. Ballard¹, Joshua Bambrick², Sebastian W. Bodenstein¹, David A. Evans¹, Chia-Chun Hung², Michael O'Neill¹, David Reiman¹, Kathryn Tunyasuvunakool¹, Zachary Wu¹, Akvilė Žemgulytė¹, Eirini Arvaniti³, Charles Beattie³, Ottavia Bertolli³, Alex Bridgland³, Alexey Cherepanov⁴, Miles Congreve⁴, Alexander I. Cowen-Rivers³, Andrew Cowie³, Michael Figurnov³, Fabian B. Fuchs³, Hannah Gladman³, Rishub Jain³, Yousuf A. Khan^{3,5}, Caroline M. R. Low⁴, Kuba Perlin³, Anna Potapenko³, Pascal Savy⁴, Sukhdeep Singh³, Adrian Stecula⁴, Ashok Thillaisundaram³, Catherine Tong⁴, Sergei Yakneen⁴, Ellen D. Zhong^{3,6}, Michal Zielinski³, Augustin Židek³, Victor Bapst^{1,8}, Pushmeet Kohli^{1,8}, Max Jaderberg^{2,8}, Demis Hassabis^{1,2,8} & John M. Jumper^{1,8}

So, What Do We Do?



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Thermodynamics of Folding

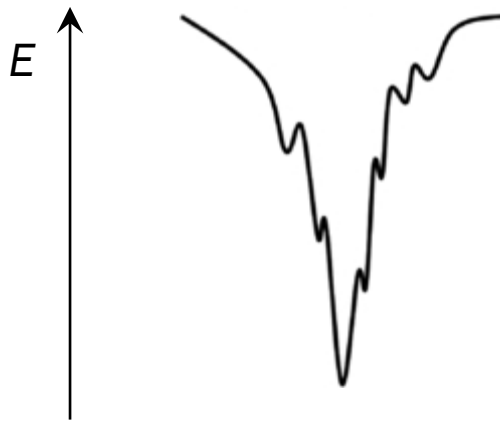


Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

Image (bottom): Mészáros et al., *Phys Biol* (2011)

What If ...

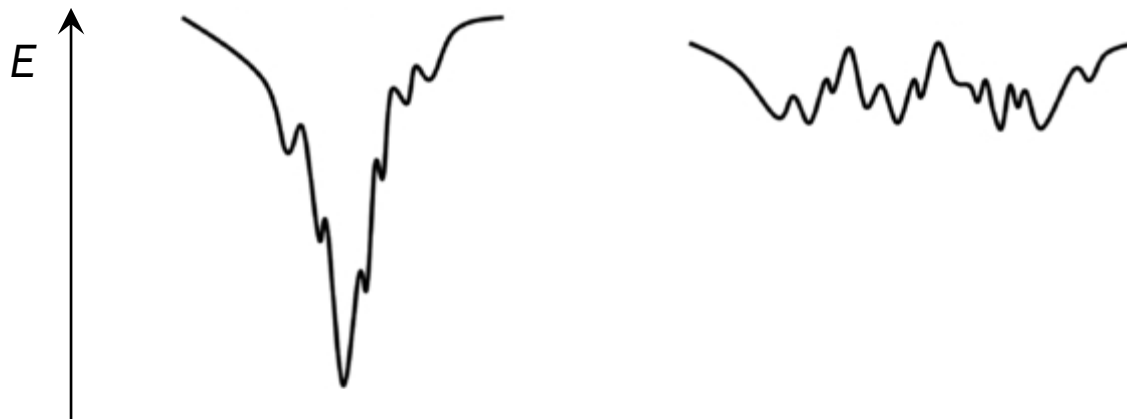


Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

Image (bottom): Mészáros et al., *Phys Biol* (2011)

No Fixed Structure

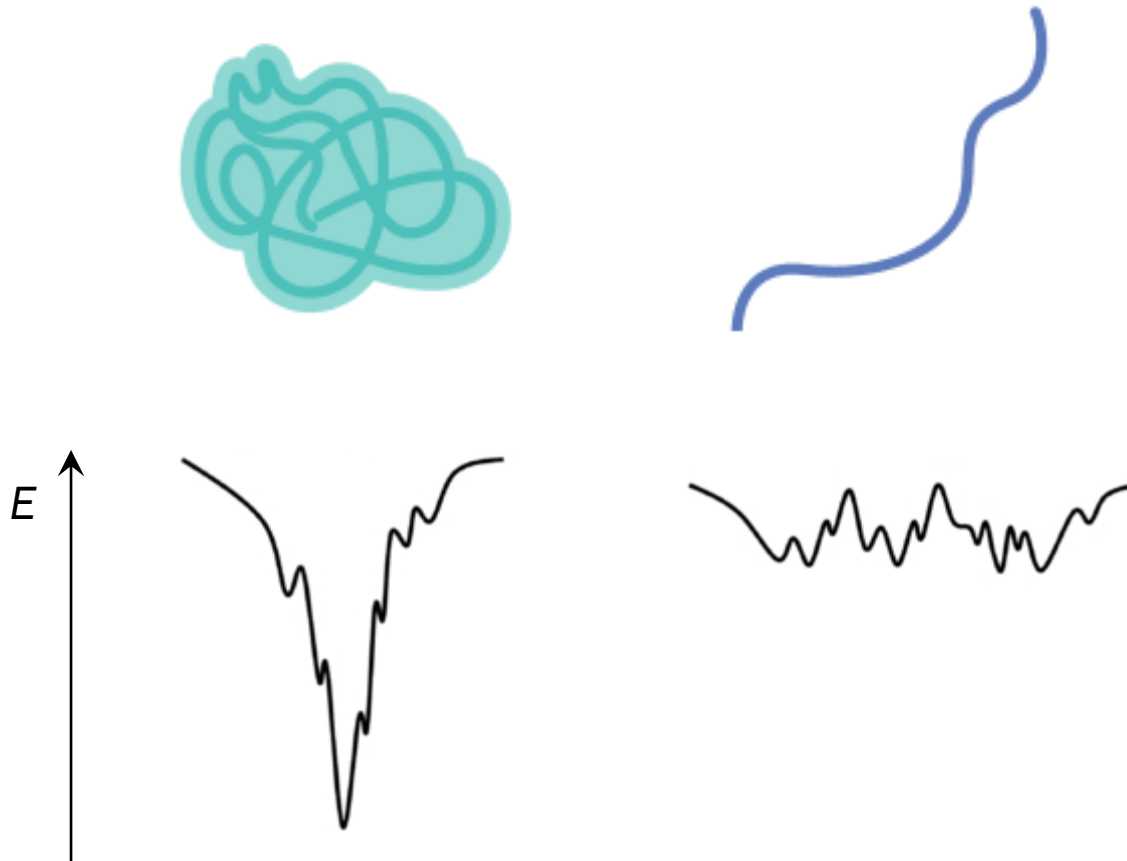


Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

Image (bottom): Mészáros et al., *Phys Biol* (2011)

Intrinsically Disordered Protein (IDP)



foldable protein



intrinsically disordered protein

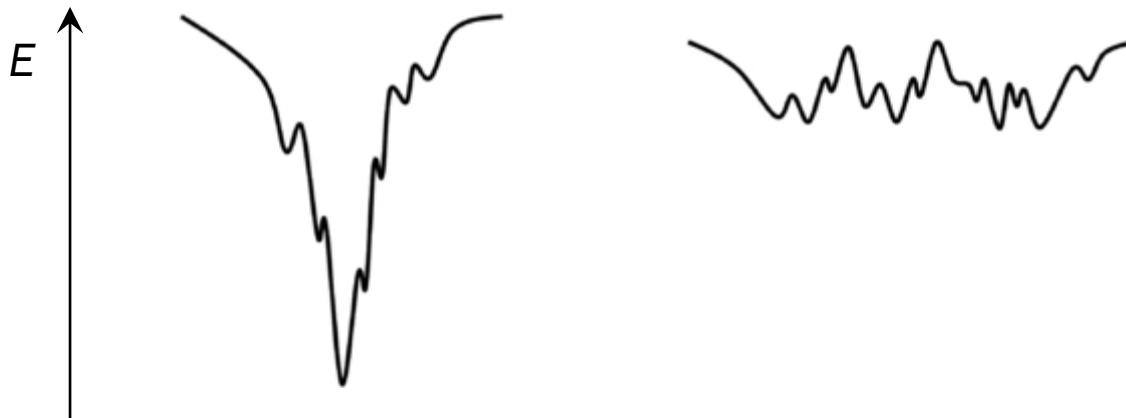
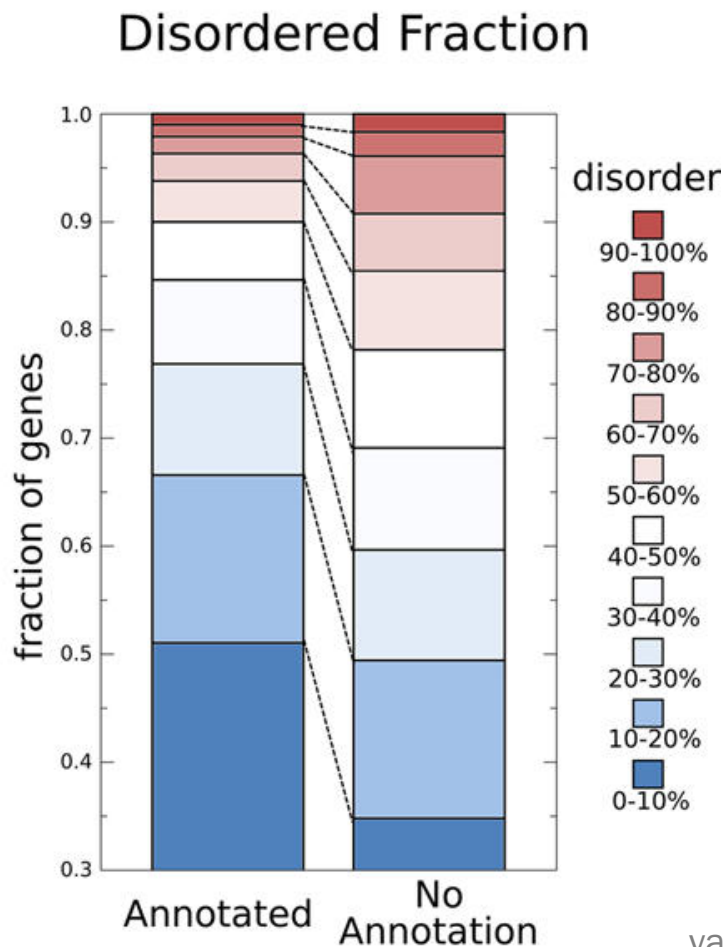


Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

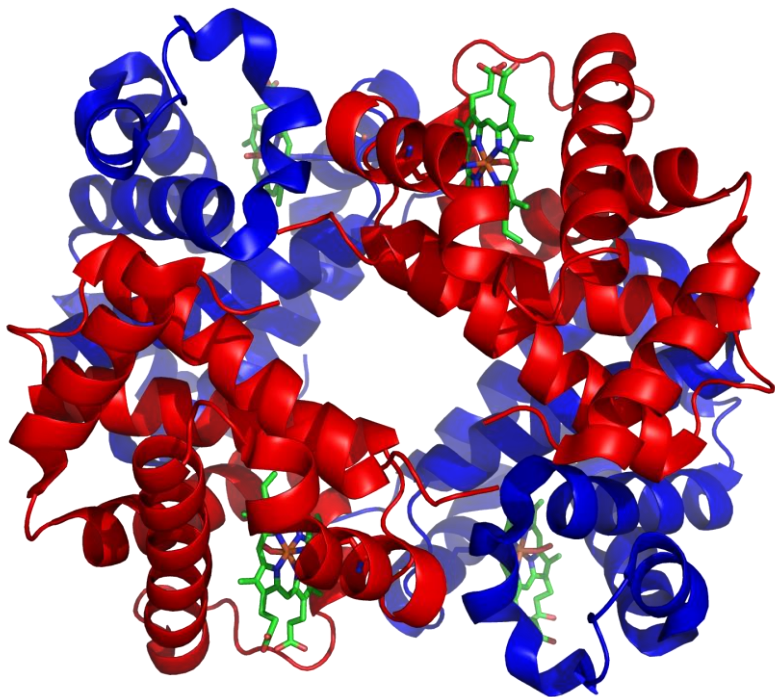
Image (bottom): Mészáros et al., *Phys Biol* (2011)

IDPs are Prevalent *in vivo*



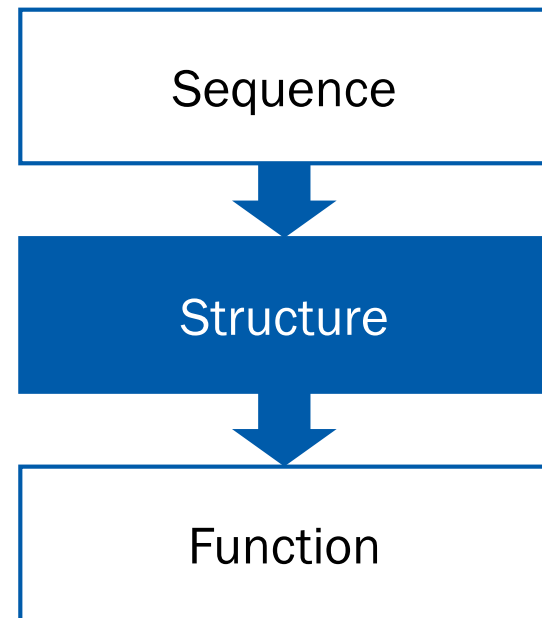
van der Lee *et al.*, *Chem Rev* (2014)

Functions of “Foldable” Proteins



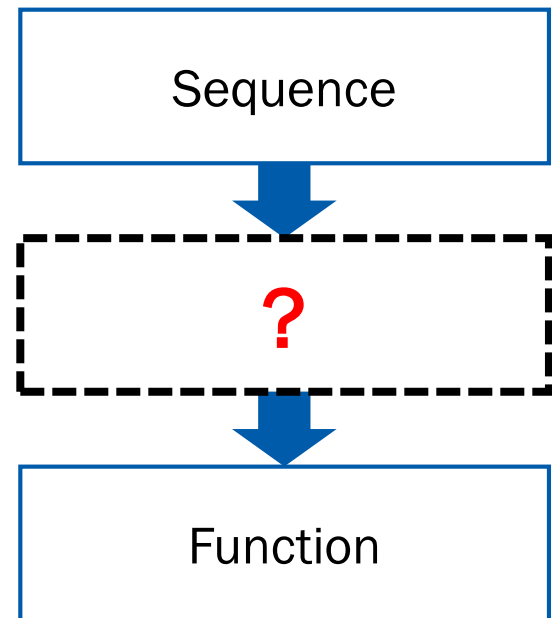
Hemoglobin

Structure-function paradigm



How do IDPs Obtain Functions?

Structure-function paradigm (?)



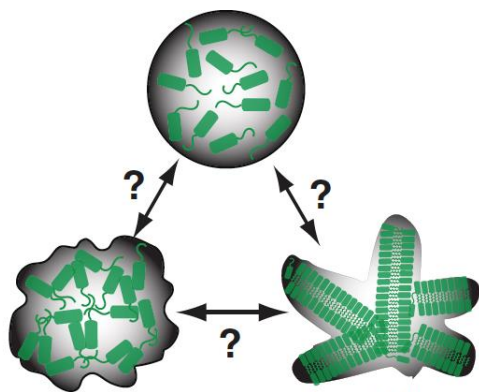
Topics



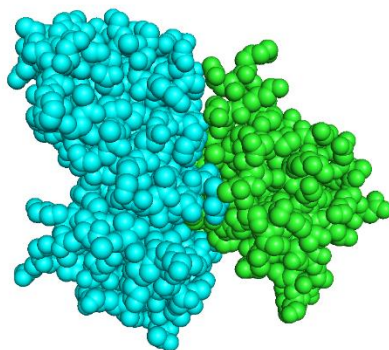
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Tools

Phase behaviors



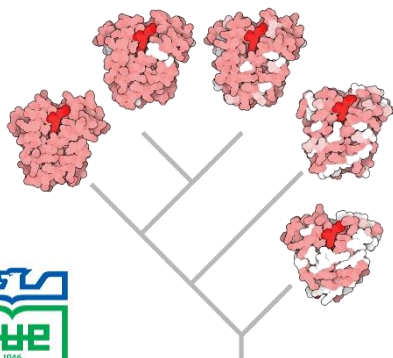
Interactions



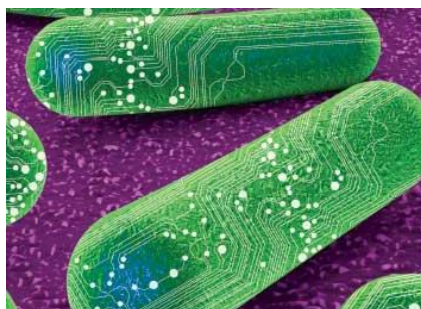
Analytics and numerics

$$\binom{N_{A \text{ stickers}}}{2N_{AA}} \times (2N_{AA} - 1)!! + \binom{N_{B \text{ stickers}}}{2N_{BB}} \times (2N_{BB} - 1)!! + \binom{N_{AB \text{ stickers}}}{N_{AB}} \times (2N_{AB} - 1)!!$$

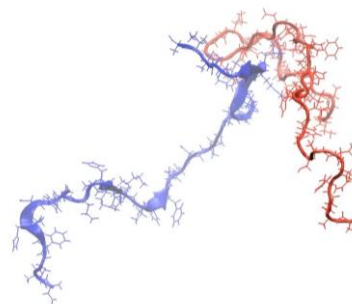
Evolution & design



Synthetic biology



Molecular simulations



in vitro experiments

